Developmental Biology

Contraction-induced endocardial *id2b* plays a dual role in regulating myocardial contractility and valve formation

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eLife Assessment

This study presents a **valuable** finding that the biomechanical force of heart contractility is required for robust endocardial id2b expression, which in return promotes valve development and myocardial function through upregulation of Neuregulin 1. The data were collected and analyzed using **solid** methodology and can be used as a starting point for deeper mechanistic insights into the genetic programs regulating endocardial-myocardial crosstalk during heart development.

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Abstract

Biomechanical cues play an essential role in sculpting organ formation. Comprehending how cardiac cells perceive and respond to biomechanical forces is a biological process with significant medical implications that remains poorly understood. Here we show that biomechanical forces activate endocardial *id2b* (inhibitor of DNA binding 2b) expression, thereby promoting cardiac contractility and valve formation. Taking advantage of the unique strengths of zebrafish, particularly the viability of embryos lacking heartbeats, we systematically compared the transcriptomes of hearts with impaired contractility to those of control hearts. This comparison identified *id2b* as a gene sensitive to blood flow. By generating a knockin reporter line, our results unveiled the presence of *id2b* in the endocardium, and its expression is sensitive to both pharmacological and genetic perturbations of contraction. Furthermore, *id2b* loss-of-function resulted in progressive heart malformation and early lethality. Combining RNA-seq analysis, electrophysiology, calcium imaging, and echocardiography, we discovered profound impairment in atrioventricular (AV) valve formation and defective excitation-contraction coupling in *id2b* mutants. Mechanistically, deletion of *id2b* reduced AV endocardial cell proliferation and led to a progressive increase in retrograde blood flow. In the myocardium, *id2b* directly interacted with the bHLH component *tcf3b* (transcription factor 3b) to restrict its activity. Inactivating *id2b* unleashed its inhibition on *tcf3b*, resulted in enhanced repressor activity of *tcf3b*, which subsequently suppressed the expression of *nrg1* (neuregulin 1), an essential mitogen for heart development. Overall, our findings identify *id2b* as an endocardial cell-specific, biomechanical signaling-sensitive gene, which mediates intercellular communications between endocardium and myocardium to sculpt heart morphogenesis and function.

Introduction

The heart develops with continuous contraction, and biomechanical cues play an essential role in cardiac morphogenesis[1,2]. Blood flow is directly sensed by the surrounding endocardium, which undergoes multiscale remodeling during zebrafish heart development. In the atrioventricular canal (AVC) endocardium, oscillatory flow promotes valvulogenesis through transient receptor potential (TRP) channel-mediated expression of Krüppel-like factor 2a (*klf2a*)[3,4,5]. Meanwhile, mechanical forces trigger ATP-dependent activation of purinergic receptors, inducing expression of nuclear factor of activated T cells 1 (*nfatc1*) and subsequent valve formation[6]. In the chamber endocardium, blood flow induces endocardial cells to adopt chamber- and region-specific cell morphology during cardiac ballooning[7]. A recent study further emphasized that blood flow is essential for endocardial cell accrual in assembling the outflow tract[8]. Beyond their role in endocardial cells, proper biomechanical cues are indispensable for shaping the myocardium. For instance, in contraction compromised *tnnt2a*[9] and *myh6*[10] mutants, trabeculation is markedly reduced. Moreover, apart from the tissue-scale regulatory effect, the shape changes[11] and myofibril content[12] at the single-cardiomyocyte level are also sculpted by the interplay of contractility and blood flow in the developing heart.

In ventricular myocardium morphogenesis, biomechanical forces coordinate intra-organ communication between endocardial and myocardial cells by regulating BMP, Nrg/Erbb, and Notch signaling. The Nrg-Erbb axis stands as one of the most extensively studied signaling pathway mediating cell-cell communications in the heart[13,14,15,16]. In particular, endocardial Notch activity induced by cardiac contraction promotes the expression of Nrg, which then secretes into the extracellular space, binding to Erbb2/4 receptor tyrosine kinases on cardiomyocyte and promoting their delamination[17,18,19]. In agreement with the pivotal role of this signaling pathway in heart development, genetic mutations in zebrafish *nrg2a* and *erbb2* result in severely compromised trabeculae formation[20,21,22].

The development of cardiomyocytes encompasses the specification of subcellular structure, metabolic state, gene expression profile, and functionality[23,24]. The rhythmic contraction of cardiomyocytes relies on precisely regulated excitation-contraction coupling (E-C coupling), transducing electrical activity into contractile forces. This intricate signaling cascade involves membrane action potential, calcium signaling, and sarcomeric structure[25,26]. Specifically, membrane depolarization triggers the opening of L-type calcium channel (LTCC), allowing calcium influx. The calcium signaling then activates the ryanodine receptor on the sarcoplasmic reticulum (SR) membrane, releasing additional calcium[25]. E-C coupling is essential for heart development, as evidenced by the complete silence of the ventricle and reduced cardiomyocyte number in *cacna1c* (LTCC α1 subunit in zebrafish) mutant[27]. Beyond its role in modulating cardiac structure formation, previous studies indicate that Nrg-Erbb2 signaling is also necessary for cardiac function, as *erbb2* mutants exhibit severely compromised fractional shortening and an immature conduction pattern[20, 28].

Id (inhibitor of DNA binding) proteins belong to the helix-loop-helix (HLH) family and function as transcriptional repressors[29]. Notably, Id2 lacks a DNA-binding domain and forms heterodimer with other bHLH proteins, acting in a dominant-negative manner[30]. Id2 plays a crucial role in heart development, and its genetic deletion results in severe cardiac defects in mice[31, 32, 33]. In zebrafish, *id2a* and *id2b* are homologs of the mammalian *Id2* gene. However, their expression pattern and function in the zebrafish heart remains largely unknown. In the present study, we identified that *id2b* is specifically expressed in endocardial cells of the developing heart, and its expression is regulated by cardiac contraction and blood flow. Genetic deletion of *id2b* led to impaired AV valve formation and reduced cardiac contractility. Therefore, *id2b* serves as a crucial mediator linking biomechanical cues to heart morphogenesis.

Results

Transcriptome analysis identifies *id2b* as a blood flow sensitive gene

Blood circulation is dispensable for early embryonic development in zebrafish, presenting an ideal model to investigate biomechanical cues influencing heart morphogenesis. To identify genes affected by cardiac contraction or blood flow, we treated *myl7:mCherry* zebrafish embryos with tricaine to inhibit cardiac contractility from 72 hours post-fertilization (hpf) to 96 hpf. Hearts from control and tricaine-treated zebrafish embryos were manually collected under a fluorescence stereoscope as previously reported[34]. Subsequently, approximately 1,000 hearts from each group were subjected to RNA sequencing (**Figure 1A**). A total of 4,530 genes with differential expression were identified, comprising 2,013 up-regulated and 2,517 down-regulated genes. With a specific focus on identifying key transcription factors (TFs) affected by perturbing biomechanical forces, differentially expressed genes (DEGs) encoding TFs were enriched into signaling pathways through KEGG analysis.

Interestingly, our analysis identified several pathways known to be involved in heart development, including the transforming growth factor beta (TGFβ) signaling and Notch signaling pathways (**Figure 1B**). In particular, the scaled expression levels of the top 6 DEGs ($|\log_2 FC| \geq 0.585$), exhibiting up- or down-regulation, were listed (**Figure 1C**). Intriguingly, *Id2* has been shown to regulate murine AV valve formation, a process notably influenced by alterations in blood flow directionality. Moreover, loss of *Id2* leads to malformation of both the arterial and venous poles of the heart and disrupts AV valve morphogenesis[31, 33]. Therefore, we interrogated the expression and function of *id2b* in developing embryos.

Quantitative real-time PCR (qRT-PCR) analysis of purified embryonic hearts revealed a significant reduction in *id2b* mRNA levels and an increase in *id2a* levels following tricaine treatment from 72 to 96 hpf compared to controls (**Figure 1D**). Furthermore, in situ hybridization was performed to visualize *id2b* expression under tricaine or 10 μM blebbistatin (an inhibitor of sarcomeric function and cardiac contractility) treatment from 48 to 72 or from 72 to 96 hpf as previously described[5].

Consistently, our results showed a reduction in *id2b* signal in contraction-deficient hearts compared to the control (**Figure 1E**). In cardiomyocytes, *tnnt2a* encodes a key sarcomeric protein essential for contractility. Similarly, injection of a previously characterized *tnnt2a*

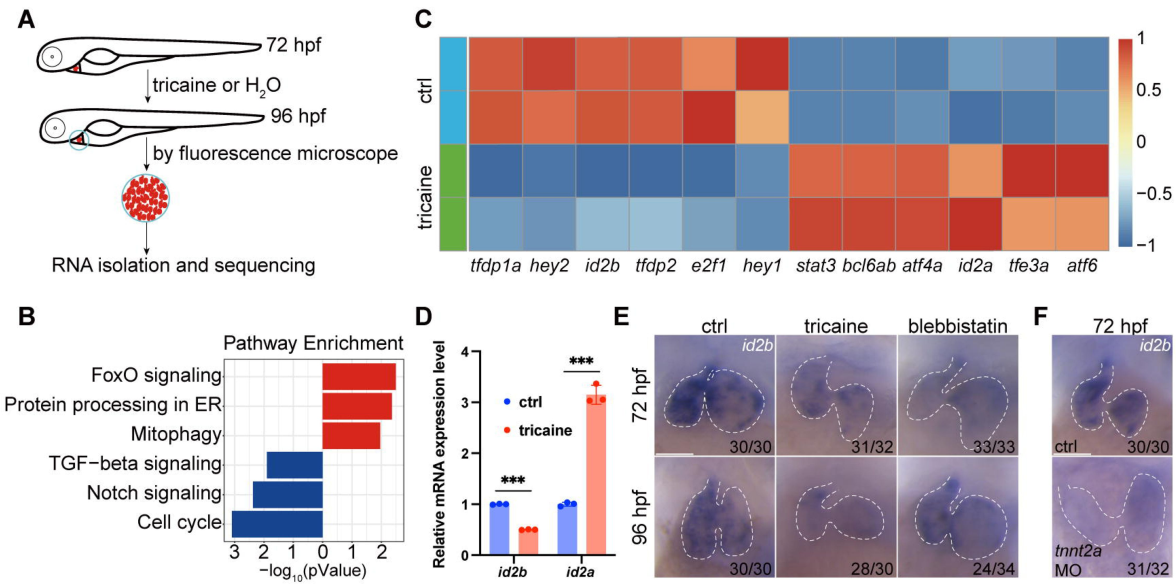

Figure 1.

Identification of *id2b* as a blood flow sensitive gene.

(**A**) Schematic showing the experimental procedures, including treatment, heart collection, and RNA-sequencing of zebrafish embryonic hearts. (**B**) KEGG enrichment analysis depicting differentially expressed genes encoding transcription factors and transcriptional regulators between control (ctrl) and tricaine-treated embryonic hearts. Red and blue rectangles represent up-regulated and down-regulated gene sets, respectively. $|\log_2$fold change$|\geq 0.585$, adjusted *p*-value<0.1. Each replicate contains approximately 1,000 hearts. (**C**) Heat map exhibiting representative genes from KEGG pathways mentioned in (**B**). (**D**) qRT-PCR analysis of *id2b* and *id2a* mRNA in control and tricaine-treated embryonic hearts. Data were normalized to the expression of *actb1*. Each sample contains ~1,000 embryonic hearts. N=three biological replicates. (**E**) *In situ* hybridization of *id2b* in 72 hpf and 96 hpf ctrl, tricaine (1 mg/mL), and blebbistatin (10 μM)-treated embryos. Numbers at the bottom of each panel indicate the ratio of representative staining. (**F**) *In situ* hybridization showing reduced *id2b* expression in *tnnt2a* morpholino-injected embryos at 72 hpf compared to control. ***$p<0.001$. Scale bars, 50 μm.

morpholino[35] at the 1-cell stage also led to compromised contraction and diminished expression of *id2b* (**Figure 1F**). Taken together, these results indicate that biomechanical cues are essential for activating *id2b* in embryonic hearts.

Visualization of the spatiotemporal expression of *id2b* in developing embryos

Due to technical challenges in visualizing the cell-type-specific expression of *id2b* in the developing heart using whole-mount in situ hybridization, we employed an intron targeting-mediated approach[36] to generate a knockin *id2b:eGFP* reporter line. This method allowed us to achieve specific labeling without perturbing the integrity and function of the endogenous gene[36] (**Figure 2A**). Comparison of *id2b:eGFP* fluorescence with in situ hybridization at 24, 48, and 72 hpf revealed that the reporter signal closely recapitulates the endogenous *id2b* expression pattern. The fluorescence was notably enriched in the heart, brain, retina, and notochord (**Figure 2B**), mirroring observations from a previously reported *id2b* transgenic line generated through BAC-mediated recombination[37].

To further elucidate the spatiotemporal expression of *id2b* in developing hearts, we crossed *id2b:eGFP* with *myl7:mCherry* or *kdrl:mCherry*, labelling cardiomyocytes or endocardial cells, respectively. Confocal images revealed minimal, if any, presence of *id2b:eGFP* in *myl7:mCherry*$^+$ cardiomyocytes (**Figure 2C**). In sharp contrast, clear co-localization between *id2b:eGFP* and *kdrl:mCherry* was evident at 48, 72, and 96 hpf (**Figure 2D**). Endocardial localization of *id2b* was further confirmed by RNAscope analysis (**Figure 2E**). In adult hearts, *id2b:eGFP* fluorescence was enriched in the chamber endocardium and the endothelium lining AVC, outflow tract (OFT), and bulbus arteriosus (**Figure 2F**). Interestingly, there was an absence of *id2b:eGFP* signal in *kdrl:mCherry*$^+$ endothelial cells in trunk blood vessel and brain vasculature (**Figure 2G**). Collectively, these results indicate that *id2b* is expressed in endocardial cells across different developmental stages.

BMP signaling and cardiac contraction regulate *id2b* expression

Taking advantage of live imaging on developing embryos, we explored the *in vivo* dynamics of *id2b* in response to biomechanical force at single-cell resolution. When embryos were treated with tricaine or blebbistatin, the intensity of *id2b:eGFP* in atrial and ventricular endocardium was significantly reduced (**Figure 3A**, B). Similarly, injection of *tnnt2a* morpholino also markedly suppressed *id2b:eGFP* signal (**Figure 3A**, B), in agreement with the results obtained from in situ hybridization. Strikingly, the reduction in fluorescence intensity was particularly pronounced in AVC endothelial cells (**Figure 3A**, B, asterisks).

We then explored how cardiac contraction modulated *id2b* expression. Given that endocardial cells can sense blood flow through primary cilia[38, 39], we used a characterized morpholino[38] to knockdown *ift88*, an intraflagellar transporter essential for primary cilia formation. Previously work demonstrated a complete loss of primary cilia in endocardial cells upon *ift88* knockdown[38]. As expected, a significant decrease in *id2b:eGFP* intensity was observed in the chamber and AVC endocardium of *ift88* morphants compared to control (**Figure 3C**, D), suggesting that biomechanical forces promote the expression of *id2b* via primary cilia. In the developing heart, a central hub for mediating biomechanical cues is the Klf2 gene, which includes the *klf2a* and *klf2b* paralogues in zebrafish[3, 4, 5, 38, 40]. Previous studies in mammals and zebrafish have highlighted the essential role of Klf2 transcription factor activity in cardiac valve and myocardial wall formation[3, 40]. As a flow- responsive gene, *klf2a* expression has been observed throughout the entire endocardium, evidenced by mRNA expression and transgenic studies[3, 4]. Interestingly, in situ hybridization on 48 and 72 hpf *klf2a*$^{-/-}$and *klf2b*$^{-/-}$ embryos unveiled a drastic decrease in *id2b* expression compared with wild-type zebrafish (**Figure 3E**), supporting the notion that *klf2*-mediated biomechanical signaling is essential for activating *id2b* expression.

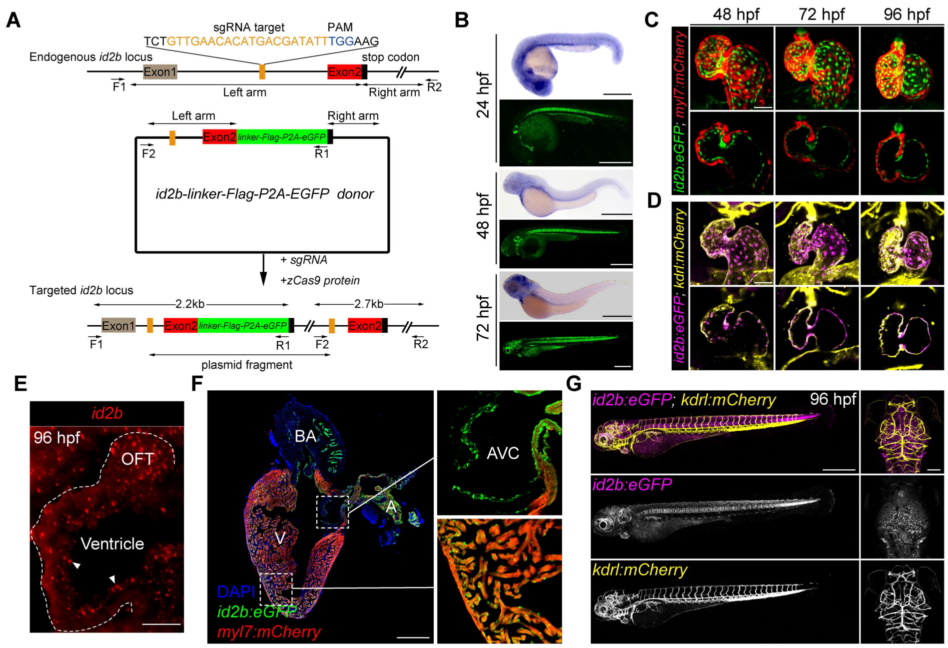

Figure 2.

The spatiotemporal expression of *id2b*.

(**A**) Schematic of the intron targeting-mediated *eGFP* knockin at the *id2b* locus using the CRISPR-Cas9 system. The sgRNA targeting sequence and the protospacer adjacent motif (PAM) sequence are shown in orange and blue, respectively. The donor plasmid comprises left and right arm sequences and a *linker-FLAG-P2A-eGFP* cassette denoted by black lines with double arrows and green box, respectively. The *linker-FLAG-P2A-eGFP* cassette was integrated into the *id2b* locus upon co-injection of the donor plasmid with sgRNA and zCas9 protein, enabling detection by PCR using two pairs of primers (F1, R1 and F2, R2)-the former length yielding a length of about 2.2 kb and the latter about 2.7 kb. (**B**) Zebrafish *id2b* expression pattern, as indicated by in situ hybridization of embryos at designated time points, was consistent with the fluorescence localization of *id2b:eGFP*, revealing expression in the heart, brain, retina, notochord, pronephric duct, and other tissues. (**C**) Maximum intensity projections (top) and confocal sections (bottom) of *id2b:eGFP; Tg(myl7:mCherry)* hearts at designated time points. (**D**) Maximum intensity projections (top) and confocal sections (bottom) of *id2b:eGFP; Tg(kdrl:mCherry)* embryos at specific time points. Magenta, *id2b:eGFP*; yellow, *kdrl:mCherry*. (**E**) RNAscope analysis of *id2b* in 96 hpf embryonic heart. White dashed line outlines the heart. OFT, outflow tract. (**F**) Immunofluorescence of adult *id2b:eGFP; Tg(myl7:mCherry*) heart section (left panel). Enlarged views of boxed areas are shown in the right panel. Green, eGFP; red, mCherry; blue, DAPI. BA, bulbus arteriosus; V, ventricle; A, atrium. AVC, atrioventricular canal. (**G**) Confocal z-stack maximum intensity projection of *id2b:eGFP;Tg(kdrl:mCherry*) embryos at 96 hpf showing the whole body (lateral view) and the head (top view). Scale bars, 500 µm (**B**, **F**, left and **G**), 50 µm (**C** and **D**), 25 µm (**E**), 100 µm (**G**, right).

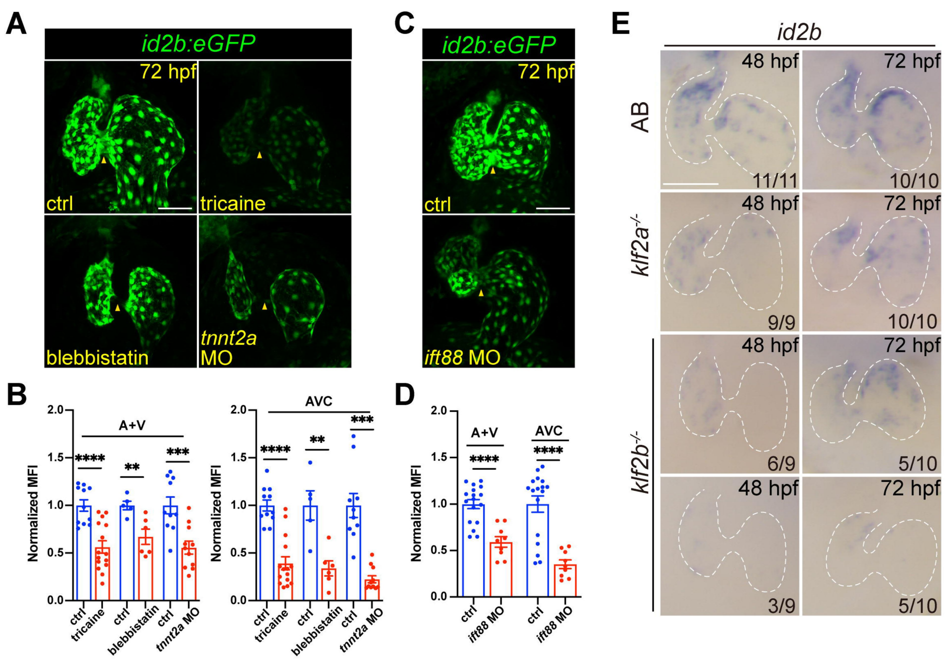

Figure 3.

Cardiac contraction promotes endocardial *id2b* expression through primary cilia.

(**A**) Representative confocal z-stack (maximal intensity projection) of *id2b:eGFP* embryos under different conditions: control, tricaine-treated, blebbistatin-treated, and *tnnt2a* morpholino-injected. Images were captured using the same magnification. (**B**) Quantification of mean fluorescence intensity (MFI) of *id2b:eGFP* in the working myocardium (atrium and ventricle, A+V) and atrioventricular canal (AVC) in (**A**). Data normalized to the mean fluorescence intensity of control hearts. n=(11, 15) (ctrl versus tricaine); n=(5, 6) (ctrl versus blebbistatin); n=(10,11) (ctrl versus *tnnt2a* MO). (**C**) Representative confocal z-stack (maximal intensity projection) of control and *ift88* morpholino-injected *id2b:eGFP* embryos. (**D**) Normalized MFI of *id2b:eGFP* in the working myocardium (A+V) and AVC in (**C**). n=(17, 9). (**E**) Whole-mount *in situ* hybridization showing *id2b* expression in control, *klf2a*$^{-/-}$, and *klf2b*$^{-/-}$embryos at 48 hpf and 72 hpf. Numbers at the bottom of each panel indicate the ratio of representative staining. Data are presented as mean ± sem. $^{**}p<0.01$, $^{***}p<0.001$, $^{****}p<0.0001$. Scale bars, 50 μm.

Given that *id2b* has been reported as a target gene of bone morphogenetic protein (BMP) signaling, we explored whether BMP also played a role in regulating *id2b* expression. To this end, we knocked down *bmp2b*, *bmp4*, and *bmp7a* in 1-cell stage embryos. Live imaging at 24 hpf revealed a significant reduction in *id2b:eGFP* fluorescence signal in morpholino-injected hearts compared to controls (Figure 3-figure supplement 1A, B), suggesting that *id2b* is a target gene of BMP signaling during early embryonic development. Similarly, treatment with the BMP inhibitor Dorsomorphin from 10 to 24 hpf resulted in a marked decrease in *id2b:eGFP* signal (Figure 3-figure supplement 1C, D). Considering that heartbeats in zebrafish commence at approximately 22 hpf, we treated embryos with Dorsomorphin from 24 to 48 hpf or from 36 to 60 hpf. While the number of endocardial cells was slightly reduced upon Dorsomorphin exposure as previously reported[7], surprisingly, quantification of the average *id2b:eGFP* fluorescence intensity in individual endocardial cells revealed no significant differences between Dorsomorphin and DMSO-treated controls (Figure 3-figure supplement 1C, D).

We further visualized BMP activity using the *BRE:d2GFP* reporter line. Confocal images revealed strong fluorescence in the myocardium at 72 hpf, with minimal signal present in the endocardium except for the AVC endothelium (Figure 3-figure supplement 1E). Moreover, after tricaine treatment, endocardial *BRE:d2GFP* slightly increased (Figure 3-figure supplement 1E), as opposed to the reduced *id2b:eGFP* signal (**Figure 3A, B**). Likewise, endocardial *BRE:d2GFP* intensity was barely affected after completely blocking contraction with *tnnt2a* MO injection (Figure 3-figure supplement 1E). These observations align with previous work using pSmad-1/5/8 as a readout of BMP activity, indicating that endocardial BMP signaling is independent of blood flow[7]. Collectively, these results suggest that *id2b* expression is regulated by both BMP and biomechanical signaling, with the relative contribution of each pathway varying across developmental stages.

Compromised AV valve formation in *id2b* mutants

To investigate the role of the contractility-*id2b* axis in zebrafish heart development, we generated a loss-of-function mutant line using CRISPR/Cas9. A pair of sgRNAs designed to target exon 1 was injected with zCas9 protein into 1-cell-stage embryos. Consequently, we identified a mutant allele with a 157 bp truncation, leading to the generation of a premature stop codon (**Figure 4A**, left). In *id2b* mutants (*id2b*$^{-/-}$), the expression levels of *id2b* were significantly decreased, while *id2a* expression levels were increased compared to *id2b*$^{+/+}$ siblings (**Figure 4A, right**). The overall morphology of *id2b*$^{-/-}$remained unaltered at 72 and 96 hpf (**Figure 4B**). However, *id2b*$^{-/-}$ zebrafish experienced early lethality starting around 31 weeks post-fertilization (**Figure 4C**). Strikingly, pericardial edema was observed in 20% (9/45) of adult *id2b*$^{-/-}$ zebrafish (**Figure 4D, top**). Upon dissecting hearts from these *id2b*$^{-/-}$ zebrafish, a prominent enlargement in the atrium with a smaller ventricle was detected (**Figure 4D, bottom**), which has been characterized as cardiomyopathy in zebrafish[41], [42]. Histological analysis further revealed malformation in the AV valves of these *id2b*$^{-/-}$ mutants compared to controls (**Figure 4E, right**). Specifically, we noted that the superior and inferior leaflets were significantly thinner, comprising only 1-2 layer of cells in *id2b*$^{-/-}$ zebrafish with an enlarged atrium. This was in sharp contrast to *id2b*$^{+/+}$ zebrafish, which exhibited multilayers of cells (**Figure 4E, left**). Subsequent examination of the remaining 80% of *id2b*$^{-/-}$ zebrafish (36/45) that did not display prominent pericardial edema also revealed AV valve malformation, albeit to a lesser extent (**Figure 4E, middle**).

To further interrogate the effect of *id2b* inactivation on AV valve formation and function, we analyzed the number of AVC endothelial cells using *kdrl:nucGFP*. At 96 hpf, a reduced number of *kdrl:nucGFP*$^{+}$ cells was detected in the AVC region of *id2b*$^{-/-}$ embryos compared with *id2b*$^{+/+}$ (Figure 4-figure supplement 1A, B). In contrast, the number of atrial and ventricular endocardial cells did not differ between *id2b*$^{-/-}$ and *id2b*$^{+/+}$ (Figure 4-figure supplement 1C, D). Subsequently, we assessed hemodynamic flow by conducting time-lapse imaging of red blood cells labelled by *gata1:dsred*.

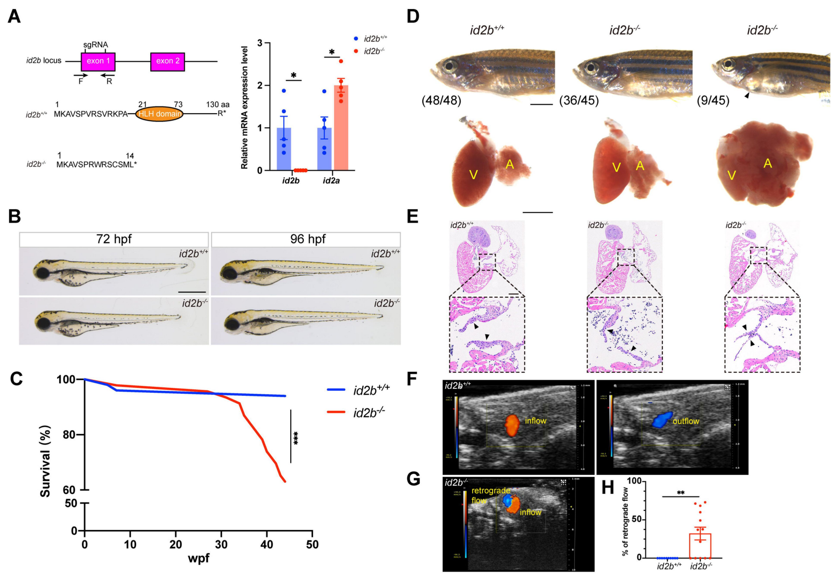

Figure 4.

id2b$^{-/-}$ adults exhibit thinner atrioventricular valve leaflets and prominent retrograde blood flow.

(**A**) Two sgRNAs, represented by short vertical lines, were designed to create *id2b*$^{-/-}$ mutants. Co-injection of the two sgRNAs with zCas9 protein induces a 157 bp truncation in the exon 1 of *id2b*, which can be detected by genotyping primers marked with arrows. This genetic modification leads to the formation of a premature stop codon and the subsequent loss of the HLH domain. Right, qRT-PCR analysis of *id2b* and *id2a* mRNA levels in *id2b*$^{+/+}$ and *id2b*$^{-/-}$ adult hearts. (**B**) No discernible morphological differences were observed between *id2b*$^{+/+}$ and *id2b*$^{-/-}$ larvae at both 72 hpf and 96 hpf. (**C**) Kaplan-Meier survival curve analysis and logrank test of *id2b*$^{+/+}$ (n=50) and *id2b*$^{-/-}$ (n=46). Wpf, weeks post-fertilization. (**D**) Pericardial edema and an enlarged atrium are evident in a subset of *id2b*$^{-/-}$ adults. V, ventricle; A, atrium. (**E**) *id2b*$^{-/-}$ adults developed thinner atrioventricular valve leaflets (denoted by arrowheads) compared to *id2b*$^{+/+}$. Enlarged views of boxed areas are shown in the bottom panels. (**F**, **G**) Echocardiograms of adult *id2b*$^{+/+}$ (**F**) and *id2b*$^{-/-}$ (**G**) hearts. Unidirectional blood flow was observed in the *id2b*$^{+/+}$ heart, while retrograde blood flow was evident in the *id2b*$^{-/-}$ heart. (**H**) Ratio of retrograde flow area over inflow area shows a significant increase in retrograde flow in *id2b*$^{-/-}$ (n=13) compared to *id2b*$^{+/+}$ (n=10). Data are presented as mean ± sem. *p<0.05, **p<0.01. Scale bars, 500 µm (**B** and **D**, bottom), 2 mm (**D**, top), 200 µm (**E**).

Surprisingly, the pattern of hemodynamics was largely preserved in *id2b*$^{-/-}$ embryos compared to *id2b*$^{+/+}$ siblings at 96 hpf (Figure 4-figure supplement 1E, Video 1, 2), suggesting that the reduced number of endocardial cells in the AVC region was not sufficient to induce functional defects.

Additionally, we performed echocardiography to analyze blood flow in adult zebrafish as previously described[43]. In *id2b*$^{-/-}$ hearts, prominent retrograde blood flow was detected in the AVC region (8/13) (**Figure 4G**), while unidirectional blood flow was observed in *id2b*$^{+/+}$ (10/10) (**Figure 4F**).

Quantification analysis showed ~32% retrograde blood flow in *id2b*$^{-/-}$, compared to 0% in *id2b*$^{+/+}$ zebrafish (**Figure 4H**). Consistently, the superior and inferior leaflets were much thinner in *id2b*$^{-/-}$ exhibiting retrograde flow compared with control fish (Figure 4-figure supplement 1F). Overall, these histological and functional analyses indicate that *id2b* deletion leads to progressive defects in AV valve morphology and hemodynamic flow.

id2b deletion perturbs calcium signaling and contractile function in the myocardium

Although similar defects in AV valve formation have been reported in both *klf2a* and *nfatc1* mutants, they do not display noticeable pericardial edema at the adult stage, nor do they experience early lethality[3, 38, 40, 43, 44]. Therefore, we sought to investigate whether other cardiac properties have also been affected by *id2b* loss-of-function. To this end, we employed RNA-seq analysis on purified embryonic *id2b*$^{-/-}$ and *id2b*$^{+/+}$ hearts (Figure 5-figure supplement 1). As expected, enrichment analysis of DEGs demonstrated that the top-ranked anatomical structures affected by *id2b* deletion included the heart valve, the compact layer of ventricle, and the atrioventricular canal (Figure 5-figure supplement 1A).

Interestingly, *id2b* inactivation also impacted phenotypes such as cardiac muscle contraction and heart contraction (Figure 5-figure supplement 1B). Therefore, we investigated cardiac contractile function through time-lapse imaging on the *myl7:mCherry* background. At 72 and 120 hpf, a significant decrease in cardiac function was observed in *id2b*$^{-/-}$compared with *id2b*$^{+/+}$ (**Figure 5A-C**, Figure 5-figure supplement 2A-C). Similarly, echocardiography analysis showed that the contractile function in adult *id2b*$^{-/-}$ heart was dramatically reduced compared with age-matched *id2b*$^{+/+}$ (**Figure 5D**, F). These functional defects in *id2b*-deleted hearts could not be attributed to differences in cardiomyocyte number, as we counted cardiomyocytes using the *myl7:H2A-mCherry* line and found no apparent changes between *id2b*$^{-/-}$ and *id2b*$^{+/+}$ embryos at 72 and 120 hpf (Figure 5-figure supplement 2D, E). Similarly, *id2b*$^{-/-}$ also developed regular trabecular structures (Figure 5-figure supplement 2F). Through α-actinin immunostaining, we observed similar sarcomeric structures in *id2b*$^{-/-}$ and control cardiomyocytes at 72 hpf and adult stages (115 dpf) (Figure 5-figure supplement 2G), corroborating that the reduced contractility in *id2b*-depleted heart was independent of structural defects.

The key functional unit that transmits electrical activity to contractile function is E-C coupling. Because *id2b*$^{-/-}$ displayed reduced cardiac function, we visualized calcium signaling in the developing heart using *actb2:GCaMP6s* zebrafish (**Figure 5G**). Compared to *id2b*$^{+/+}$ controls, *id2b*$^{-/-}$ embryos exhibited markedly decreased calcium transient amplitude (**Figure 5H**), consistent with compromised calcium handling observed in other zebrafish cardiomyopathy models[42, 45]. In cardiomyocyte, the entry of extracellular calcium is mainly mediated through the LTCC. As previously reported, a defect in zebrafish LTCC pore forming α1 subunit *cacna1c* leads to compromised cardiac function[27]. We collected hearts from 72 hpf and 5 months post-fertilization (mpf) zebrafish and detected downregulated *cacna1c* in *id2b*$^{-/-}$ compared to *id2b*$^{+/+}$ (**Figure 5I**). In addition, we measured cardiac action potential using intracellular recording[46]. Compared to *id2b*$^{+/+}$ zebrafish, the duration of the action potential in *id2b*$^{-/-}$was significantly

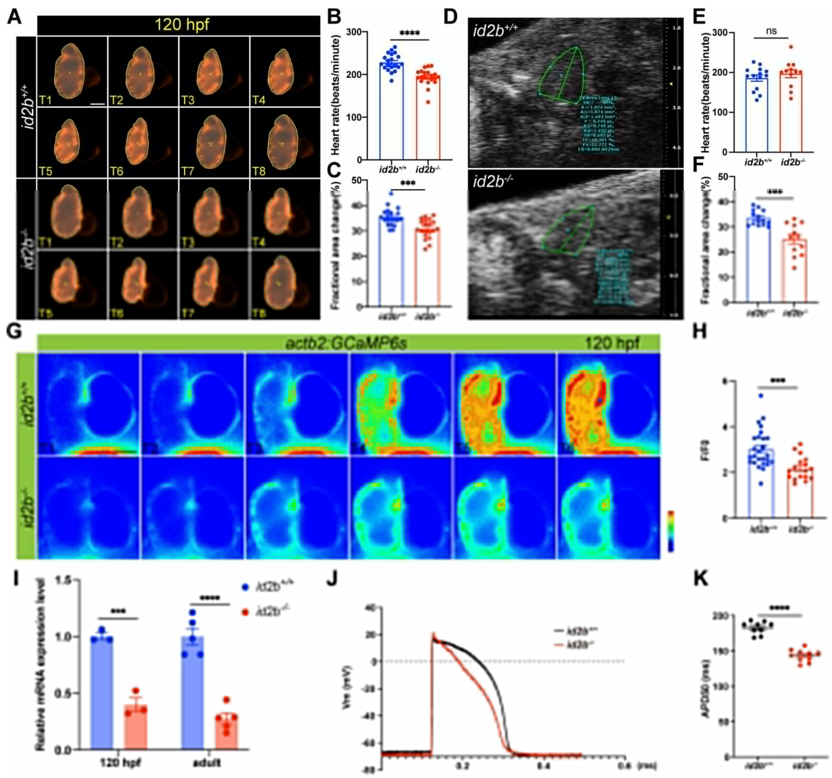

Figure 5.

Reduced cardiac contractile function and compromised calcium handling in *id2b*$^{-/-}$ mutants.

(**A**) Time-lapse imaging (from T1 to T8) illustrates the cardiac contraction-relaxation cycle of 120 hpf *id2b*$^{+/+}$ and *id2b*$^{-/-}$ hearts carrying *myl7:mCherry*. (**B** and **C**) *id2b*$^{-/-}$ larvae (n=20) display a significant decrease in heart rate and fractional area change compared to *id2b*$^{+/+}$ (n=20). (**D**) Echocardiograms of adult *id2b*$^{+/+}$ and *id2b*$^{-/-}$ hearts. (**E** and **F**) *id2b*$^{-/-}$ fish (n=12) exhibit reduced cardiac contractile function with preserved heart rate compared to *id2b*$^{+/+}$ (n=14). (**G**) Time-lapse imaging illustrates the calcium dynamics of 120 hpf *id2b*$^{+/+}$ and *id2b*$^{-/-}$ hearts carrying *actb2:GCaMP6s*. (**H**) Ratio of maximal fluorescence intensity (F) over basal fluorescence intensity (F0) of GCaMP6s signal. n=(26, 17). (**I**) qRT-PCR analysis of *cacna1c* mRNA in *id2b*$^{+/+}$ and *id2b*$^{-/-}$ hearts at 120 hpf (N=three biological replicates, with each sample containing 500-1,000 embryonic hearts) and adult stage (N=five biological replicates). Data were normalized to the expression of *actb1*. (**J**) The action potential of ventricular cardiomyocyte in adult *id2b*$^{+/+}$ (n=9) and *id2b*$^{-/-}$ (n=9) hearts. (**K**) Statistical data showed a notable difference between *id2b*$^{+/+}$ and *id2b*$^{-/-}$ accordingly. Data are presented as mean ± sem. ***p<0.001, **** p<0.0001, ns, not significant. Scale bars, 50 μm.

shorter (**Figure 5J**, K), consistent with the decreased expression level of *cacna1c*. Together, these data indicate that *id2b* loss-of-function leads to compromised calcium signaling and cardiac contractile function.

Reduced expression of *nrg1* mediates the compromised contractility in *id2b*$^{-/-}$

Because the deficiency of *id2b* in the endocardium disrupted the function of myocardium, we speculated that the crosstalk between these two types of cells was affected in *id2b*$^{-/-}$. Interestingly, comparing the differentially expressed genes in embryonic *id2b*$^{-/-}$ and *id2b*$^{+/+}$ hearts identified a significant reduction in the expression level of Nrg1, a key mitogen regulating the intra-organ communications between endocardial cells and cardiomyocytes (**Figure 6A**). Remarkably, analysis of a zebrafish single-cell database[47] revealed enriched expression of *nrg1* in endocardial cells (Figure 6-figure supplement 1). However, attempts to detect *nrg1* expression through in situ hybridization were unsuccessful, likely due to its low abundance in the heart. Alternatively, qRT-PCR analysis of purified 120 hpf embryonic hearts validated decreased *nrg1* levels in *id2b*$^{-/-}$ compared to control (**Figure 6B**). Previous studies have demonstrated that perturbations in Nrg-Erbb2 signaling, as seen in zebrafish *erbb2* mutants, result in dysfunctional cardiac contractility[20]. Consistently, a decrease in heart rate was observed in embryos treated with the *erbb2* inhibitor AG1478 (**Figure 6C**).

Remarkably, injecting *nrg1* mRNA at the 1-cell stage not only rescued the reduced expression of *cacna1c* in *id2b*$^{-/-}$ hearts (**Figure 6D**) but also restored the diminished heart rate (**Figure 6E**). This is consistent with prior study showing that Nrg1 administration can restore LTCC expression and calcium current in failing mammalian cardiomyocytes[48]. Overall, our data suggest that endocardial *id2b* promotes Nrg1 synthesis, thereby enhancing cardiomyocyte contractile function.

Id2b interacts with Tcf3b to limit its repressor activity on *nrg1* expression

We further interrogated how *id2b* promotes the expression of *nrg1*. As a HLH factor lacking a DNA binding motif, Id2b has been reported to form a heterodimer with Tcf3b to limits its function as a potent transcriptional repressor[49]. Notably, we detected expression of *tcf3b* in endocardial cells by analyzing a zebrafish single-cell database[47] (Figure 7-figure supplement 1). To determine if zebrafish Id2b and Tcf3b interact *in vitro*, Flag-*id2b* and HA-*tcf3b* were co-expressed in HEK293 cells. Co- immunoprecipitation analysis confirmed their interaction (**Figure 7A**), although whether they interact *in vivo* remains to be further investigated. Subsequently, qRT-PCR analysis on purified 120 hpf embryonic hearts revealed a significant increase in the expression of *socs3b* and *socs1a*, target genes of *tcf3b*, in *id2b*$^{-/-}$ compared to *id2b*$^{+/+}$ (**Figure 7B**). This suggests an elevation in *tcf3b* activity associated with the loss of *id2b* function. Notably, the expression levels of *tcf3a* and *tcf3b* remained consistent between *id2b*$^{-/-}$ and *id2b*$^{+/+}$ hearts (**Figure 7B**).

To understand how the altered interaction between *id2b* and *tcf3b* influences *nrg1* expression, we analyzed the promoter region of zebrafish *nrg1* using JASPAR and identified two potential *tcf3b* binding sites (**Figure 7C**). Subsequently, a DNA fragment containing the zebrafish *nrg1* promoter region was subcloned into a vector carrying the luciferase reporter gene. Co-injection of this construct with *tcf3b* mRNA into 1-cell stage embryos resulted in a significant decrease in luciferase signal.

Conversely, co-injection with a previously characterized *tcf3b* morpholino led to enhanced luciferase intensity (**Figure 7D**). These results suggest a possible mechanism by which Tcf3b represses *nrg1* expression in zebrafish.

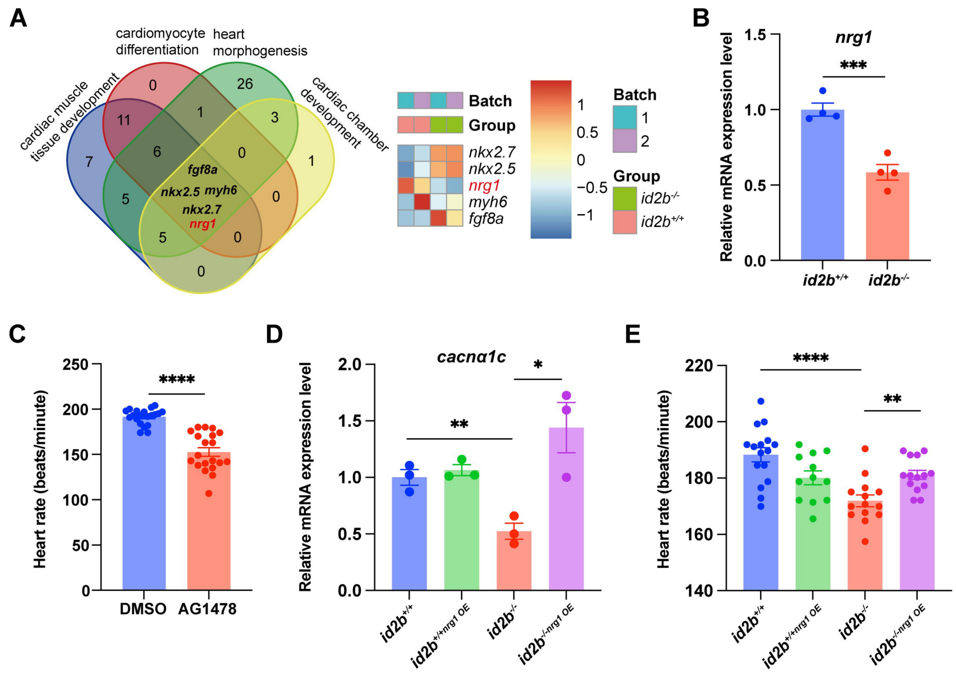

Figure 6.

Nrg1 serves as a pivotal mitogen mediating the function of *id2b*.

(**A**) Identification of genes (*fgf8a*, *nkx2.5*, *myh6*, *nrg1*, and *nkx2.7*) associated with four distinct heart development processes: cardiac muscle tissue development, cardiomyocyte differentiation, heart morphogenesis, and cardiac chamber development. The heatmap illustrates scaled-normalized expression values for the mentioned genes. (**B**) qRT-PCR analysis of *nrg1* mRNA in 120 hpf *id2b*$^{+/+}$ and *id2b*$^{-/-}$ embryonic hearts. Data were normalized to the expression of *actb1*. N=four biological replicates, with each sample containing 500-1,000 embryonic hearts. (**C**) Heart rate in 120 hpf larvae treated with AG1478 (n=20) and DMSO (n=20). (**D**) *id2b*$^{+/+}$ and *id2b*$^{-/-}$ larvae were injected with *nrg1* mRNA at the 1-cell stage, followed by qRT-PCR analysis of *cacnα1c* mRNA at 72 hpf. Data were normalized to the expression of *actb1*. N=three biological replicates, with each sample containing 100-200 embryonic hearts. (**E**) The heart rate of 72 hpf *id2b*$^{+/+}$ and *id2b*$^{-/-}$ larvae injected with *nrg1* mRNA at 1-cell stage. n=(16, 12, 14, 14). Data are presented as mean ± sem. **p*<0.05, ***p*<0.01, ****p*<0.001, *****p*<0.0001. ns, not significant.

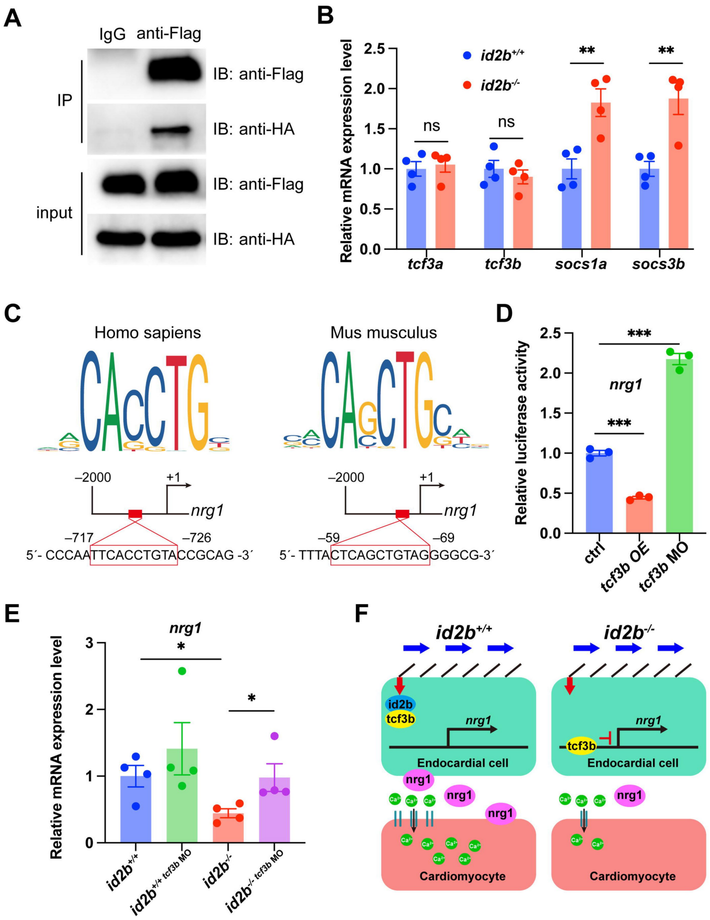

Figure 7.

Id2b interacts with Tcf3b to restrict its inhibition on *nrg1* expression.

(**A**) Immunoprecipitation (IP) assays of Flag-*id2b* and HA-*tcf3b* co-transfected 293T cells. (**B**) qRT-PCR analysis of *tcf3a*, *tcf3b*, *socs1a*, and *socs3b* mRNA in 120 hpf *id2b*$^{+/+}$ and *id2b*$^{-/-}$ embryonic hearts. Data were normalized to the expression of *actb1*. N=four biological replicates, with each sample containing 500-1,000 embryonic hearts. (**C**) Two potential Tcf3b binding sites, with sequences corresponding to the human TCF3 (left) and mouse Tcf3 (right) binding motifs, were predicted in the 2,000bp DNA sequence upstream of the zebrafish *nrg1* transcription start site using JASPAR. (**D**) Luciferase assay showing the expression of *nrg1* in embryos with *tcf3b* overexpression (*tcf3b* OE) and morpholino-mediated *tcf3b* knockdown (*tcf3b* MO). N=three biological replicates. (**E**) qRT-PCR analysis of *nrg1* mRNA in 72 hpf *id2b*$^{+/+}$ and *id2b*$^{-/-}$ embryonic hearts injected with control and *tcf3b* morpholino. Data were normalized to the expression of *actb1*. N=four biological replicates, with each sample containing 100-200 embryonic hearts. (**F**) Schematic model for *id2b*-mediated regulation of myocardium function. During heart development, blood flow operates through primary cilia, initiating endocardial *id2b* expression. Subsequently, the interaction between Id2b and Tcf3b restricts the activity of Tcf3b, ensuring proper *nrg1* expression, which in turn promotes LTCC expression (left). However, in the absence of Id2b, Tcf3b inhibits *nrg1* expression. The reduced Nrg1 hinders LTCC expression in cardiomyocytes, resulting in decreased extracellular calcium entry and disruption of myocardial function. Data are presented as mean ± sem. *p<0.05, **p<0.01, ***p<0.001. ns, not significant.

Lastly, injecting *tcf3b* morpholino into *id2b*$^{-/-}$ embryos was performed to assess whether attenuating the overactive *tcf3b* in *id2b*$^{-/-}$ could restore the expression level of *nrg1*. qRT-PCR analysis of purified 72 hpf hearts revealed a partial restoration of the diminished *nrg1* expression in *id2b*$^{-/-}$ upon *tcf3b* inhibition (**Figure 7E**). Taken together, our results indicate that biomechanical cues activate endocardial Id2b expression, leading to its interaction with Tcf3b to alleviate repression on the *nrg1*promoter. Consequently, the depletion of *id2b* unleashes Tcf3b's repressor activity, leading to a reduction in *nrg1* expression, which further acts through *erbb2* to regulate cardiomyocyte function (**Figure 7F**).

Discussion

Biomechanical forces play an essential role in regulating the patterning and function of the heart. At AVC, oscillatory flow promotes the expression of *klf2a* and *nfatc1* to modulate valve morphogenesis. In chamber endocardium, blood flow induces endocardial cells to acquire distinct cell morphology. However, it still lacks a systematic analysis of the transcriptome underlying compromised heartbeats. In the present study, we analyzed embryonic zebrafish hearts without contractility and identified genes that are regulated by biomechanical forces. Specifically, our results unveiled the endocardial-specific expression of *id2b*, which was tightly regulated by flow-sensitive primary cilia-*klf2* axis. Genetic deletion of *id2b* resulted in compromised valve formation and progressive atrium enlargement. In addition, a reduction in heart rate and contractile force was observed in *id2b*$^{-/-}$, owing to decreased expression of LTCC α1 subunit *cacna1c*. Mechanistically, *id2b* interacts with bHLH transcription factor *tcf3*b to limit its repressor activity. Hence genetic deletion of *id2b* unleashes *tcf3*b activity, which further represses endocardial *nrg1* expression. As a result, Injection of *nrg1* mRNA partially rescues the phenotype of *id2b* deletion. Overall, our findings identify *id2b* as a novel mediator that regulate the interplay between endocardium and myocardium during heart development.

In mammals, the deletion of *Id2* leads to malformations in the arterial and venous poles of the heart, as well as affects AV valve morphogenesis[31, 33]. Interestingly, approximately 20% perinatal lethality is reported in *Id2* knockout mice, exhibiting AV septal defects and membranous ventricular septal defects[33]. Remarkably, pericardial edema is evident in 20% of adult *id2b*$^{-/-}$ zebrafish, with a prominent enlargement of the atrium. The superior and inferior leaflets of AV valves in *id2b*$^{-/-}$ mutants are significantly thinner compared to the control. Therefore, our results suggest that *id2b* may play a similar role in regulating AV valve formation in zebrafish as its mammalian orthologue *Id2*. It is proposed that the loss of *Id2* in mice results in compromised endocardial proliferation and aberrant endothelial-to mesenchymal transformation (EMT), collectively leading to defective valve morphogenesis[33]. Nevertheless, the mechanism by which *id2b* loss-of-function causes a reduction in leaflet thickness in zebrafish remains to be determined in future studies.

id2b has been recognized as a target gene of the BMP signaling pathway. As expected, knockdown of *bmp2b*, *bmp4*, and *bmp7a* at 1-cell stage confirms that endocardial *id2b* expression is controlled by BMP activity during early embryonic development. Surprisingly, treatment with the BMP inhibitor Dorsomorphin at 24 and 36 hpf, when cardiac contractions have already initiated, fails to alter *id2b* expression in the endocardium, suggesting that BMP is dispensable for *id2b* activation at these stages. Instead, endocardial *id2b* expression is reduced upon loss-of-function of *klf2a*, *klf2b*, and *ift88*, suggesting an essential role of the primary cilia-*klf2* axis in mediating *id2b* activation. In endocardial cells, Trp, Piezo, and ATP-dependent P2X/P2Y channels[4, 6, 50, 51] are well-established sensors for biomechanical stimulation. The activation of these channels further promotes the activities of Klf2 and Nfatc1 to drive heart development and valvulogenesis. However, whether these channels are also required for the activation of *id2b* warrants further investigation.

The Nrg-Erbb signaling plays an essential role in regulating heart morphogenesis and function. In the mammalian heart, the genetic deletion of Nrg1 or Erbb2 results in severely perturbed cardiac trabeculae formation[13], [14], [15]. Zebrafish *erbb2* mutants exhibit a similar defect in cardiomyocyte proliferation and trabeculation[20]. Interestingly, *nrg1* mutant zebrafish display grossly normal cardiac structure during early embryonic devolpment[21], [52]. Nevertheless, zebrafish *nrg2a* loss-of-function leads to defective trabeculae formation, suggesting that *nrg2a* is the predominant ligand secreted from endocardium, promoting ventricular morphogenesis[21]. In the adult stage, perivascular cells[53] or regulatory T cells[54]-derived *nrg1* promotes cardiomyocyte proliferation during heart regeneration.

Hence, the specific ligand/receptor and the spatiotemporal regulation of the Nrg-Erbb axis appear to be more complicated in both embryonic and adult zebrafish. Interestingly, the *nrg1* mutant heart exhibits a defect in cardiac nerve expansion and heart maturation at the juvenile stage despite normal cardiac structure formation[52], suggesting its potential role in regulating cardiac function. Our findings demonstrate that the expression of *nrg1* in embryonic endocardial cells is influenced by biomechanical cues and *id2b* activity. This signaling axis is essential for coordinating endocardium-myocardium interaction and establishing proper cardiac function.

Materials and methods

Zebrafish handling and lines

All animal procedures were approved by the Animal Care and Use Committee of the Zhejiang University School of Medicine (application no.29296). Embryonic and adult fish were raised and maintained under standard conditions at 28 °C on a 14/10 hour day/night cycle. The following zebrafish lines were used in this study: *Tg(myl7:mCherry)*^{sd7}[55], *Tg(myl7:H2A-mCherry)*^{sd12}[56], *Tg(kdrl:mCherry)*^{S896}[57], *Tg(kdrl:nucGFP)*^{y7}[58], *Tg(BRE:d2GFP)*^{mw30}[59], and *Tg(actb2:Gcamp6s)*. To generate the *id2b* mutant, two short guide RNAs (sgRNAs) targeting exon 1 were generated using the MAXIscript T7 transcription kit (ambion, AM1314). The sgRNAs were as follows: sgRNA1: 5’ - GAAGGCAGTCAGTCCGGTG - 3’; sgRNA2: 5’ - GAACCGGAGCGTGAGTAAGA - 3’. The two sgRNAs, along with zCas9 protein, were co-injected into one-cell stage embryos. Embryos were raised to adulthood and crossed to wild-type zebrafish to obtain F_1 progenies. Through PCR analysis, a mutant line with a 157 bp truncation was identified.

The knock-in *id2b:eGFP* line was generated using a previously reported method[36]. Briefly, three sgRNAs were designed to target the intron of *id2b* (sgRNA1: 5’ - GAGACAAATATCTACTAGTG - 3’; sgRNA2: 5’ - GTTGAACACATGACGATATT - 3’; sgRNA3: 5’ - GCACAACTTAGATTTCAAGT -3’). Co-injection of each individual sgRNA with zCas9 protein into one-cell stage zebrafish embryos yielded varying cleavage efficiency. Since sgRNA2 displayed the highest gene editing efficiency, it was selected for subsequent studies. Next, a donor plasmid containing the sgRNA targeting sequence of the intron, exon 2 of *id2b*, and P2A-eGFP was generated. Co-injection of sgRNA, donor plasmid, and zCas9 protein into one-cell stage embryos led to concurrent cleavage of the sgRNA targeting sites in both the zebrafish genome and the donor plasmid (**Figure 2A**). Accordingly, eGFP fluorescence was observed in injected 24 hpf zebrafish embryos, indicating the incorporation of the donor. The insertion of the *id2b*-p2A-eGFP donor into the genome was confirmed by PCR analysis with primers recognizing target site or donor sequences, respectively (**Figure 2A**). Embryos with mosaic eGFP expression were raised to adulthood and crossed with wild-type zebrafish to obtain F_1 progenies.

Overall, two founders were identified. The junction region of the F_1 embryos was sequenced to determine the integration sites. Although the two founders had slightly different integration sites in the intron, the expression pattern and fluorescence intensity of eGFP were indistinguishable between the two lines.

Morpholinos

All morpholinos (Gene Tools) used in this study have been previously characterized. *tnnt2a* MO (5' - CATGTTTGCTCTGATCTGACACGCA - 3')[35]; *ift88* MO (5' - CTGGGACAAGATGCACATTCTCCAT - 3')[38]; *bmp2b* MO (5' - ACCACGGCGACCATGATCAGTCAGT - 3')[60]; *bmp4* MO (5' - AACAGTCCATGTTTGTCGAGAGGTG - 3')[61]; *bmp7a* MO (5' - GCACTGGAAACATTTTTAGAGTCAT - 3')[60]; *tcf3b* MO (5' -CGCCTCCGTTAAGCTGCGGCATGTT - 3')[62]. For each morpholino, a 1 nL solution was injected into one-cell stage embryos at the specified concentrations: 0.5 µg/µL *tnnt2a* MO, 2 µg/µL *ift88* MO, 0.5 µg/µL *bmp2b* MO, 2 µg/µL *bmp4* MO, 4 µg/µL *bmp7a* MO, and 1 µg/µL *tcf3b* MO.

Small molecules treatment

To inhibit cardiac contraction, embryos were incubated in 1 mg/mL tricaine (Sigma, A5040) or 10 µM blebbistatin (MedChemExpress, HY13441) PTU-added egg water for 12-24 hours. In order to inhibit *erbb2* signaling pathway, 10 µM AG1478 (Sigma, 658552) were used to treat 4 dpf larvae. To inhibit BMP signaling pathway, 10 µM Dorsomorphin (Sigma, P5499) was used to treat 10, 24, and 36 hpf embryos.

In situ hybridization and RNAscope

Whole mount *in situ* hybridization was performed as previously described[46]. The probes were synthesized using the DIG RNA labeling kit (Roche). The primers used for obtaining the *id2b* probe template were as follows: Forward 5' - ATGAAGGCAGTCAGTCCGGTGAGGT - 3'; Reverse 5' - TCAACGAGACAGGGCTATGAGGTCA - 3'. RNAscope analysis was performed using the probe Dr- *id2b* (Advanced Cell Diagnostics, 517541) and the Multiplex Fluorescent Detection Kit version 2 (Advanced Cell Diagnostics, 323100) as previously described[19].

Embryonic heart isolation and RNA-seq analysis

Hearts were isolated from embryos carrying the *Tg*(*myl7:mCherry)* transgene following an established protocol[34]. A minimum of 1,000 hearts for each experimental group were manually collected under a Leica M165FC fluorescence stereomicroscope and transferred into ice-cold PBS buffer. After centrifugation at 12, 000 g for 2 min at 4°C, the supernatant was removed, and hearts were lysed in cold Trizol buffer (Ambion, 15596). Total RNA was extracted for subsequent quantitative real-time PCR or RNA-seq analysis.

Duplicate samples from control and Tricaine-treated embryonic hearts underwent RNA-seq. Raw sequencing reads were preprocessed to remove adapters and filter low-quality reads. Clean sequencing reads were then mapped to the zebrafish reference[63] using STAR with default parameters[64].

Subsequently, gene quantification was carried out with RSEM[65]. The gene expected count was applied to identify differential expression genes (DEGs), retaining only genes with counts per million (CPM) of 10 in at least two samples. DESeq2[66] was employed for differential expression analysis, and *P*-values were adjusted using BH correction. DEGs were defined as those with $|\log_2$ fold change$| \geq 0.585$ and an adjusted *P*-value < 0.1. The primary focus was on genes related to transcription regulation, and gene enrichment analysis was conducted using ClusterProfiler[67]. To analyze DEGs in *id2b*$^{-/-}$ and control embryonic hearts, we performed enrichment analysis with the R package EnrichR, dissecting the potential anatomy expression pattern and underlying phenotypes. We mainly focused on genes with the heart related phenotypes, including cardiac muscle tissue development, cardiomyocyte differentiation, heart morphogenesis and cardiac chamber development. All the analysis on identifying DEGs were batch corrected.

Quantitative real-time PCR analysis

After extraction from isolated embryonic hearts, 50 ng-1 ug of mRNA was reverse transcribed to cDNA using the PrimeScript RT Master Mix kit (Takara, RR036A). Real-time PCR was performed using the TB Green Premix Ex Taq kit (Takara, RR420A) on the Roche LightCycler 480. Expression levels of the target genes were normalized to *actb1* as an internal control. All experiments were repeated three times. The following primer sets were used: *id2b* Forward 5' - ACCTTCAGATCGCACTGGAC - 3', Reverse 5' - CTCCACGACCGAAACACCATT - 3'; *nrg1* Forward 5' - CTGCATCATGGCTGAGGTGA - 3', Reverse 5' - TTAACTTCGGTTCCGCTTGC - 3'; *cacna1c* Forward 5' - GCCCTTATTGTAGTGGGTAGTG - 3', Reverse 5' -AGTGTTTTGGAGGCCCATTG - 3'; *tcf3a* Forward 5' - CCTCCGGTCATGAGCAACTT - 3', Reverse 5' - TTTCCCATGATGCCTTCGCT - 3'; *tcf3b* Forward 5' - CCTTTAATGCGCCGTGCTTC -3', Reverse 5' - GCGTTCTTCCATTCCTGTACCA - 3'; *socs1a* Forward 5' - TCAGCCTGACAGGAAGCAAG - 3', Reverse 5' - GTTGCACAGGGATGCAGTCG - 3'; *socs3b* Forward 5' - GGGACAGTGAGTTCCTCCAA - 3', Reverse 5' - ATGGGAGCATCGTACTCCTG - 3'; *actb1* Forward 5' - ACCACGGCCGAAAGAGAAAT - 3', Reverse 5' - GCAAGATTCCATACCCAGGA - 3'.

Co-immunoprecipitation (Co-IP) and western blot

Zebrafish *tcf3b* and *id2b* were overexpressed in 293T cell for 48 hours. The transfected cells were then collected and lysed using IP lysis buffer (Sangon Biotech, C500035) containing protease and phosphatase inhibitors (Sangon Biotech, C510009, C500017). For the IP experiment, anti-Flag antibody (Cell Signaling Technology, 14793, 1:100) and IgG antibody (ABclonal, AC005, 1:100) was incubated with cell lysates overnight at 4°C. Pretreated magnetic beads were bound with the antigen- antibody complex for 4 hours at 4°C, followed by washing with IP lysis buffer three times. For western blot, samples were denatured at 95°C for 10 min, separated on a 5%-12% gradient gel. Proteins were then transferred to a PVDF membrane (Sigma, ISEQ00010). The membrane was blocked for 1 hour with 5% nonfat milk or 5% BSA (Sangon Biotech) dissolved in TBST and then incubated with primary antibodies (anti-FLAG, Cell Signaling Technology, 14793, 1:1000; anti-HA, Sigma, H3663, 1:1000) overnight at 4°C, followed by three times 10-min TBST washes. HRP-conjugated secondary antibodies (Invitrogen, 31430, 31460) were incubated for 1 hour at room temperature, followed by three times 10- min TBST washes. The detection of immunoreactive bands was performed with a chemiluminescent substrate (Thermo Scientific, 34577) and imaged using the Azure Biosystems 400.

Immunofluorescence

For immunofluorescence on adult zebrafish hearts, we fixed the hearts overnight in 4% paraformaldehyde at 4 L, followed by equilibration through 15% and 30% sucrose in PBS solution. The hearts were embedded and frozen in O.C.T. compound (Epredia, 6502), and 10 μm thick cryosections were prepared using a CryoStar NX50 cryostat. Immunofluorescence experiments were performed as previously described[68]. For immunofluorescence on embryonic hearts, embryos were fixed overnight in 4% paraformaldehyde at 4, washed twice quickly in 100% methanol, and then dehydrated overnight at -20L in 100% methanol. Subsequently, rehydration was performed through a methanol gradient (100%, 75%, 50%, 25%, 10 min each), followed by three times washes in PBST (1% PBS/0.1% Triton-X100, 10 min each). The embryos were treated with 10 μg/mL proteinase K diluted in PBST for 20 min at room temperature, re-fixed in 4% paraformaldehyde for 20 min, washed in PBST, and immersed in blocking solution (PBST/1% BSA/2% goat serum) for 1 hour at room temperature. Following this, the embryos were incubated in the primary antibody diluted in blocking solution overnight at 4L. After washing in PBST, they were incubated in the secondary antibody (1:200) for 2 hours at room temperature. The primary antibody used was: Anti-GFP antibody (Santa Cruz Biotechnology, sc9996, 1:200), anti-α-actinin antibody (Sigma, A7811, 1:200). The secondary antibody used was: Anti-mouse IgG-Alexa 488 (Invitrogen, A11011, 1:400). DAPI was used to stain cell nuclei.

Cardiac function analysis

To assess cardiac function in embryonic hearts, embryos were incubated in 0.16 mg/mL tricaine (Sigma, A5040) and embedded in 1% low melting agarose. Heart contractions were recorded for 1 min using a Nikon Ti2 microscope at a rate of 25 frames per second. Fractional shortening and heart rate were measured as described previously[46]. For cardiac contractile functions in adulthood, zebrafish were fixed on a sponge soaked with system water with the belly facing up and echocardiography was performed[69]. Videos and images in color Doppler mode and B-mode were obtained using the Vevo1100 imaging system at a frequency of 50 MHz. Nikon NIS elements AR analysis and Image J software were employed for data extraction. To evaluate AV valve function, the ratio of inflow and outflow area in the same frame was quantified[43].

Calcium imaging

At 120 hpf, embryos were treated with 10 mM 2,3-butanedione monoxime (BDM, Sigma, B0753) and mounted in 1% low melting agarose. Time-lapse images were acquired using a Nikon Ti2 microscope at a rate of 50 frames per second. Data were analyzed using Nikon NIS elements AR analysis software.

Intracellular action potential recording

Electrophysiology study was performed on adult zebrafish ventricles as previously described[46]. Briefly, hearts were mounted in a chamber containing Tyrode's solution: NaCl 150 mM, KCl 5.4 mM, $MgSO_4$ 1.5 mM, NaH_2PO_4 0.4 mM, $CaCl_2$ 2 mM, Glucose 10 mM, HEPES 10 mM, pH was adjusted to 7.4.

Glass pipettes with tip resistance 30-40 MΩ were filled with 3M KCl solution. Intracellular action potentials were recorded using an HEKA amplifier and pClamp10.3 software (Molecular Devices).

Histology and HE staining

Adult hearts were dissected and fixed overnight at 4°C in 4% PFA, followed by three times PBS washes. Dehydration involved an ethanol gradient (70%, 80%, 95%, 100%, 100%, 30 min each), followed by three soaks in dimethylbenzene at 65°C, before embedding in paraffin. Sections of 5 μm thickness were prepared using the Leica RM2235 manual rotary microtome for hematoxylin and eosin (HE) staining.

Injection of mRNA

The embryonic zebrafish cDNA library was used as a template to amplify the *nrg1* and *tcf3b* fragment, which was then subcloned into the pCS2 vector. The vector was linearized using Not I restriction endonuclease, and mRNA was transcribed in vitro using the mRNA transcription kit (Ambion, AM1340). 100 pg of purified mRNA was injected into one-cell stage embryos.

Luciferase assay

The LCR (luciferase reporter) plasmid was generated by subcloning the 5' UTR of *nrg1* into the upstream region of renilla luciferase on the psiCheck2 plasmid. Following construction, 25 pg of the LCR plasmid was co-injected with either 100 pg of *tcf3b* mRNA or 1 ng of *tcf3b* MO into 1-cell stage zebrafish embryos. At 48 hpf, twenty embryos were gathered into one group and fully lysed.

Subsequently, firefly and renilla luciferase activities were sequentially measured using a microplate reader with the dual luciferase reporter gene assay kit (Yeasen, 11402ES60), according to the manufacturer's instructions. The experiment was independently replicated three times. The relative renilla luciferase activity, normalized by firefly luciferase activity, served as an indicator of *nrg1* expression level under the influence of *tcf3b* overexpression or reduction.

Image processing and statistical analysis

Whole mount *in situ* hybridization images were captured with a Leica M165FC stereomicroscope. Live imaging of zebrafish embryos involved mounting anesthetized embryos in 1% low melting agarose (Sangon Biotech, A600015) and manually oriented them for optimal visual access to the heart.

Confocal images were obtained with a Nikon Ti2 confocal microscope. Fluorescence intensity and cell number counting were processed using Nikon NIS Elements AR analysis and Image J software.

Statistical analysis was performed using Graphpad prism 8 software. No statistical methods were used to predetermine sample size. Unpaired two-tailed Student's t-tests was used to determine statistical significance. Data are presented as mean ± s.e.m, $^{*}P < 0.05$ was considered to be statistically significant.

Data availability

The authors declare that all data supporting the findings in the paper are available in the article and the supplementary files. RNA-seq data have been deposited in GEO under accession number GSE295737 and GSE295738.

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Additional information

Author Contributions

Shuo Chen and Peidong Han contributed to the study design. Shuo Chen, Jinxiu Liang, Weijia Zhang, Peijun Jiang, Wenyuan Wang, Xiaoying Chen and Yuanhong Zhou conducted experiments. Jie Yin analyzed RNA-sequencing data. Peng Xia, Fan Yang, Ying Gu, and Ruilin Zhang provided key reagents and analytic tool. Shuo Chen, Jinxiu Liang, Jie Yin, and Peidong Han prepared the manuscript.

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Additional files

Figure 3—figure supplement 1. *Blood flow and BMP signaling independently activates id2b expression. (**A**) Representative confocal maximal intensity projection of control (non-injected), bmp2b, bmp4, and bmp7a morpholino-injected id2b:eGFP; Tg(myl7:mCherry) hearts at 24 hpf. White circles outline eGFP signal. (**B**) Quantification of mean fluorescence intensity of id2b:eGFP in (**A**). Data normalized to the mean fluorescence intensity of control hearts. n=(7, 9, 8, 10). (**C**) Representative confocal maximal intensity projection of DMSO and Dorsomorphin (DM)-treated id2b:eGFP hearts at 24, 48 and 60 hpf. Embryos were treated from 10 to 24 hpf, from 24 to 48 hpf, or from 36 to 60 hpf. White circles outline eGFP signal. (**D**) Quantification of mean fluorescence intensity of id2b:eGFP in (**C**). Data normalized to the mean fluorescence intensity of DMSO-treated hearts. n=(6, 10) (24 hpf); n=(14, 16) (48 hpf); n=(9, 9) (60 hpf). (**E**) Confocal optical sections of control, tricaine-treated, and tnnt2a morpholino-injected 72 hpf Tg(BRE:d2GFP); Tg(kdrl:mCherry) hearts. Yellow arrowheads, endocardial cells. Numbers at the top of each panel indicate the ratio of representative images. (**F**) Schematic diagram of blood flow and BMP signaling-mediated id2b expression. Data are presented as mean ± sem. ***p<0.001, ****p<0.0001. ns, not significant. Scale bars, 50 μm.*

Figure 4—figure supplement 1. *Figure 4-figure supplement 1. id2b$^{-/-}$ larvae exhibit a decreased number of valve endocardial cells while maintaining normal atrioventricular valve function. (**A**) Representative confocal images of valve endocardial cells in 96 hpf id2b$^{+/+}$ and id2b$^{-/-}$ hearts carrying Tg(kdrl:nucGFP). (**B**) Quantification of the number of valve endocardial cells in id2b$^{+/+}$ and id2b$^{-/-}$ hearts. VECs, valve endocardial cells. n=(10, 10). (**C**) Representative confocal images of 72 and 120 hpf id2b$^{+/+}$ and id2b$^{-/-}$ hearts in Tg(kdrl:nucGFP); Tg(myl7:mCherry) transgenic background. (**D**) Statistical analysis of the number of endocardial cells in the ventricle (V), atrium (A), and combined (A+V) in id2b$^{+/+}$ and id2b$^{-/-}$ hearts. n=(13, 13) (72 hpf); n=(12, 15) (120 hpf). (**E**) Quantification of blood flow patterns in 96 hpf id2b$^{+/+}$ (n=16) and*

id2b$^{-/-}$ (n=15) hearts. (**F**) H&E staining of adult id2b$^{-/-}$ and id2b$^{+/+}$ hearts after echocardiographic analysis in Figure 4F and 4G. Enlarged views of boxed areas are shown in the bottom panels. Data are presented as mean ± sem. ****p<0.0001. ns, not significant. Scale bars, 20 μm (**A**), 50 μm (**C**), 200 μm (**F**).

Figure 5—figure supplement 1. *id2b loss-of-function impacts both valve formation and cardiac contraction. (**A**) The top 20 predicted tissue expression patterns based on the differential expressed genes from id2b$^{+/+}$ and id2b$^{-/-}$ hearts at 120 hpf were displayed. (**B**) The top 20 predicted phenotypes related to the differential expressed genes from id2b$^{+/+}$ and id2b$^{-/-}$ hearts at 120 hpf were illustrated.*

Figure 5—figure supplement 2. *id2b$^{-/-}$ hearts develop normal trabeculae and sarcomeres. (**A**) Time-lapse imaging (from T1 to T8) illustrates the cardiac contraction-relaxation cycle of 72 hpf id2b$^{+/+}$ and id2b$^{-/-}$ hearts carrying myl7:mCherry. (**B** and **C**) Heart rate and fractional area change in id2b$^{-/-}$ (n=21) and id2b$^{+/+}$ (n=21) at 72 hpf. (**D**) Representative confocal z-stack of 72 and 120 hpf id2b$^{+/+}$ and id2b$^{-/-}$ hearts with Tg(myl7:H2A-mCherry) transgene. (**E**) Quantification of the number of cardiomyocytes in the ventricle (V), atrium (A), and combined (A+V) in (**D**). n=(8, 10) (72 hpf); n=(10, 10) (120 hpf). (**F**) Representative confocal images of 72 and 120 hpf id2b$^{+/+}$ and id2b$^{-/-}$ hearts carrying Tg(myl7:mCherry). (**G**) Confocal immunofluorescence images of α-actinin (green) in embryonic (72 hpf) and adult (115 dpf) hearts (left). Right: fluorescence intensity profiles for α-actinin. Data are presented as mean ± sem. ns, not significant. Scale bars, 50 μm (**D** and **F**), 5 μm (**G**).*

Figure 6—figure supplement 1. *nrg1 is expressed in the endocardial cells. (**A**) Analysis of nrg1 expression using the zebrafish single-cell landscape database. The red rectangle highlights endocardial cells. (**B**) Identification of the endocardial cell population in the zebrafish cell landscape. Purple dots represent endocardial cells, and the red oval denotes cell populations from the heart. Images were generated using the Zebrafish Cell Landscape (ZCL) at http://bis.zju.edu.cn/ZCL/.*

Figure 7—figure supplement 1. *Expression landscape of zebrafish tcf3b. **(A)** The major cell types expressing tcf3b were displayed according to the zebrafish single-cell landscape. **(B)** Identification of the endocardial cell population in the zebrafish cell landscape. Purple dots represent endocardial cells, and the red oval denotes cell populations from the heart. Images were generated using the Zebrafish Cell Landscape (ZCL) at http://bis.zju.edu.cn/ZCL/.*

Figure 4—figure supplement 1—video 1. *96 hpf id2b$^{+/+}$ larvae displayed unidirectional blood flow in the AV canal. Scale bar, 50 μm.*

Figure 4—figure supplement 1—video 2. *96 hpf id2b$^{-/-}$ larvae displayed unidirectional blood flow in the AV canal. Scale bar, 50 μm.*

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Joint Public Review:

Summary:

How mechanical forces transmitted by blood flow contribute to cardiac development remains incompletely understood. Using the unique advantages of the zebrafish model, Chen et al make the fundamental discovery that endocardial expression of the transcriptional repressor, Id2b, is maintained in endocardial cells by blood flow. Id1b zebrafish mutants fail to form the valve in the atrioventricular canal (AVC) and show reduced myocardial contractility that they suggest is due to impaired calcium transients. Id2b mutants are largely viable during the first 6 months of life until ~20% display cardiomyopathy characterized by visible edema, structural abnormalities, retrograde blood flow, and reduced systolic function and calcium handling. Mechanistically, the authors suggest that flow-mediated expression of Id2b leads to neuregulin 1 (nrg1) upregulation by physically interacting with and sequestering the Tcf3b transcriptional repressor from conserved tcf3b binding sites upstream of nrg1. Overall, this study advances our understanding of flow-mediated endocardial-myocardial crosstalk during heart development.

Strengths:

The strengths of the study are the significance of the biological question being addressed, use of the zebrafish model, data quality, and use of genetic tools. The text is generally well-written and easy to understand.

Weaknesses:

The main weakness that remains is the lack of rigor surrounding the molecular mechanism where the authors suggest that blood flow induces endocardial expression of Id2b, which binds to Tcf3b and sequesters it from binding the Nrg1 promoter to repress transcription. Although good faith efforts were made to bolster their model, the physical interaction between Id2b and Tcf3b is limited to overexpression of tagged proteins in HEK293 cells. Moreover, no mutagenesis was performed on the tcf3b binding sites identified in the nrg1 promoter to learn their importance in vivo.

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Author response:

The following is the authors' response to the original reviews

> ***Public Reviews:***
>
> ***Reviewer #1 (Public review):***
>
> *Summary:*
>
> *Chen et al. identified the role of endocardial id2b expression in cardiac contraction and valve formation through pharmaceutical, genetic, electrophysiology, calcium imaging, and echocardiography analyses. CRISPR/Cas9 generated id2b mutants demonstrated defective AV valve formation, excitation-contraction coupling, reduced endocardial cell proliferation in AV valve, retrograde blood flow, and lethal effects.*
>
> *Strengths:*
>
> *Their methods, data and analyses broadly support their claims.*
>
> *Weaknesses:*
>
> *The molecular mechanism is somewhat preliminary.*

We thank the reviewer for the positive assessment of our work. A detailed point-by-point response has been incorporated in the response to "Recommendations for the authors" section.

> ***Reviewer #2 (Public review):***
>
> *Summary:*
>
> *Biomechanical forces, such as blood flow, are crucial for organ formation, including heart development. This study by Shuo Chen et al. aims to understand how cardiac cells respond to these forces. They used zebrafish as a model organism due to its unique strengths, such as the ability to survive without heartbeats, and conducted transcriptomic analysis on hearts with impaired contractility. They thereby identified id2b as a gene regulated by blood flow and is crucial for proper heart development, in particular, for the regulation of myocardial contractility and valve formation. Using both in situ hybridization and transgenic fish they showed that id2b is specifically expressed in the endocardium, and its expression is affected by both pharmacological and genetic perturbations of contraction. They further generated a null mutant of id2b to show that loss of id2b results in heart malformation and early lethality in zebrafish. Atrioventricular (AV) and excitation-contraction coupling were also impaired in id2b mutants. Mechanistically, they demonstrate that Id2b interacts with the transcription factor Tcf3b to restrict its activity. When id2b is deleted, the repressor activity of Tcf3b is enhanced, leading to suppression of the expression of nrg1 (neuregulin 1), a key factor for heart development. Importantly, injecting tcf3b morpholino into id2b-/- embryos partially*

> *restores the reduced heart rate. Moreover, treatment of zebrafish embryos with the Erbb2 inhibitor AG1478 results in decreased heart rate, in line with a model in which Id2b modulates heart development via the Nrg1/Erbb2 axis. The research identifies id2b as a biomechanical signaling-sensitive gene in endocardial cells that mediates communication between the endocardium and myocardium, which is essential for heart morphogenesis and function.*
>
> *Strengths:*
>
> *The study provides novel insights into the molecular mechanisms by which biomechanical forces influence heart development and highlights the importance of id2b in this process.*
>
> *Weaknesses:*
>
> *The claims are in general well supported by experimental evidence, but the following aspects may benefit from further investigation:*
>
> *(1) In Figure 1C, the heatmap demonstrates the up-regulated and down-regulated genes upon tricane-induced cardiac arrest. Aside from the down-regulation of id2b expression, it was also evident that id2a expression was up-regulated. As a predicted paralog of id2b, it would be interesting to see whether the up-regulation of id2a in response to tricane treatment was a compensatory response to the down-regulation of id2b expression.*

We thank the reviewer for the comment. As suggested, we performed qRT-PCR analysis to assess *id2a* expression in tricaine-treated heart. Our results demonstrate a significant upregulation of *id2a* following the inhibition of cardiac contraction, suggesting a potential compensatory response to the decreased *id2b*. These new results have been incorporated into the revised manuscript (Figure 1D).

> *(2) The study mentioned that id2b is tightly regulated by the flow-sensitive primary cilia-klf2 signaling axis; however aside from showing the reduced expression of id2b in klf2a and klf2b mutants, there was no further evidence to solidify the functional link between id2b and klf2. It would therefore be ideal, in the present study, to demonstrate how Klf2, which is a transcriptional regulator, transduces biomechanical stimuli to Id2b.*

We have examined the expression levels of *id2b* in both *klf2a* and *klf2b* mutants. The whole mount in situ results clearly demonstrate a decrease in *id2b* signal in both mutants (Figure 3E). As noted by the reviewer, *klf2* is a transcriptional regulator, suggesting that the regulation of *id2b* may occur at the transcriptional level. However, dissecting the molecular mechanisms underlying the crosstalk between *klf2* and *id2b* is beyond the scope of the present study.

> *(3) The authors showed the physical interaction between ectopically expressed FLAG-Id2b and HA-Tcf3b in HEK293T cells. Although the constructs being expressed are of zebrafish origin, it would be nice to show in vivo that the two proteins interact.*

We thank the reviewer for this insightful comment. As suggested, we synthesized Flag-*id2b* and HA-*tcf3b* mRNA and co-injected them into 1-cell stage zebrafish embryos. We collected 100-300 embryos at 12, 24, and 48 hpf and performed western blot analysis using the same anti-HA and anti-Flag antibodies validated in HEK293 cell experiments. Despite multiple independent attempts, we were unable to detect clear bands of the tagged proteins in zebrafish embryos. We speculate that this could be due to mRNA instability, translational efficiency, or the low abundance of Id2b and Tcf3b proteins. We have acknowledged these technical limitations in the revised manuscript and clarified that the HEK293 cell data support a potential interaction between Id2b and Tcf3b, while confirming their endogenous interaction will require further investigations (Lines 295-296).

> ***Reviewer #3 (Public review):***
>
> *Summary:*
>
> *How mechanical forces transmitted by blood flow contribute to normal cardiac development remains incompletely understood. Using the unique advantages of the zebrafish model system, Chen et al make the fundamental discovery that endocardial expression of id2b is induced by blood flow and required for normal atrioventricular canal (AVC) valve development and myocardial contractility by regulating calcium dynamics. Mechanistically, the authors suggest that Id2b binds to Tcf3b in endocardial cells, which relieves Tcf3b-mediated transcriptional repression of Neuregulin 1 (NRG1). Nrg1 then induces expression of the L-type calcium channel component LRRC1. This study significantly advances our understanding of flow-mediated valve formation and myocardial function.*
>
> *Strengths:*
>
> *Strengths of the study are the significance of the question being addressed, use of the zebrafish model, and data quality (mostly very nice imaging). The text is also well-written and easy to understand.*
>
> *Weaknesses:*
>
> *Weaknesses include a lack of rigor for key experimental approaches, which led to skepticism surrounding the main findings. Specific issues were the use of morpholinos instead of genetic mutants for the bmp ligands, cilia gene ift88, and tcf3b, lack of an explicit model surrounding BMP versus blood flow induced endocardial id2b expression, use of bar graphs without dots, the artificial nature of assessing the physical interaction of Tcf3b and Id2b in transfected HEK293 cells, and artificial nature of examining the function of the tcf3b binding sites upstream of nrg1.*

We thank the reviewer for the positive assessment and the constructive suggestions. We have performed additional experiments and data analysis to address these issues. A detailed point-by-point response has been incorporated in the response to “Recommendations for the authors” section.

> ***Recommendations for the authors:***
>
> ***Reviewer #1 (Recommendations for the authors):***
>
> *Questions/Concerns:*
>
> *(1) In the introduction, it would be beneficial to include background information on the id2b gene, what is currently known about its function in heart development/regeneration and in other animal models than just the zebrafish.*

We thank the reviewer for the constructive suggestion. In the revised manuscript, we have added a paragraph in the Introduction to provide background on *id2b* and its role in heart development. Specifically, we discuss its function as a member of the ID (inhibitor of DNA binding) family of helix-loop-helix (HLH) transcriptional regulators and highlight its involvement in cardiogenesis in both zebrafish and mouse models. These additions help place our findings in a broader developmental and evolutionary context (Lines 91-100).

> *(2) Of the 6 differentially expressed genes identified in Figure 1C, why did the authors choose to focus on id2b and not the other 5 downregulated genes?*

We thank the reviewer for the comments. As suggested, we have added a sentence in the revised manuscript to clarify the rationale for selecting *id2b* as the focus of the present study (Lines 117-121).

> *(3) As the authors showed representative in situ images for id2b expression with blebbistatin treatment in Figure 1E, and tnn2a MO in Figure 1F, it would also be beneficial to show relative mRNA expression levels for id2b in conditions of blebbistatin treatment and tnn2a MO knockdown. In Fig. 1C: id2b is downregulated with tricaine, but id2a is upregulated with tricaine. Do these genes perform similar or different functions, results of gene duplication events?*

We thank the reviewer for the thoughtful suggestion. Our in situ hybridization results demonstrate reduced *id2b* expression following tricaine, blebbistatin, and *tnn2* morpholino treatment. To further validate these observations and enhance cellular resolution, we generated an *id2b:eGFP* knockin line. Analysis of this reporter line confirmed a significant reduction in *id2b* expression in the endocardium upon inhibition of cardiac contraction and blood flow (Figure 3A-D), supporting our in situ results. The divergent expression patterns of *id2a* and *id2b* in response to tricaine treatment likely reflect functional specification following gene duplication in zebrafish. While our current study focuses on characterizing the role of *id2b* in zebrafish heart development, the specific function of *id2a* remains to be determined.

> *(4) In Fig. 2b, could the authors compare the id2b fluorescence with RNAscope ISH at 24, 48, and 72 hpf? RNAscope ISH allows for the visualization of single RNA molecules in individual cells. The authors should at least compare these in the heart to demonstrate that id2b accurately reflects the endogenous id2b expression. In Fig. 2E: Suggest showing the individual fluorescent images for id2b:eGFP and kdrl:mCherry in the same colors as top panel images instead of in black and white. In Fig. 2F: The GFP fluorescence from id2b:eGFP signals looks overexposed.*

We thank the reviewer for the valuable comment. In response, we attempted RNAscope in situ hybridization on embryos carrying the *id2b:eGFP* reporter to directly compare fluorescent reporter expression with endogenous *id2b* transcripts. However, we encountered a significant reduction in *id2b:eGFP* fluorescence following the RNAscope procedure, and even subsequent immunostaining with anti-GFP antibodies yielded only weak signals. Despite this technical limitation, the RNAscope results independently confirmed *id2b* expression in endocardial cells (Figure 2E), supporting the specificity and cell-type localization observed with the reporter line. As suggested by the reviewer, we have updated Figure 2G to display *id2b:eGFP* and *kdrl:mCherry* images in the same color scheme as the top panel to improve consistency and clarity. Additionally, we have replaced the images in Figure 2F to avoid overexposure and better represent the spatial distribution of *id2b:eGFP* in adult heart.

> *(5) In Fig. 3A: are all the images in panel A taken with the same magnification? In Fig. 3e, could the authors show the localization of klf2 and id2b and confirm their expression in the same endocardial cells? In Fig. 3, the authors conclude that klf2-mediated biomechanical signaling is essential for activating id2b expression. This statement is somewhat overstated because they only demonstrated that knockout of klf2 reduced id2b expression.*

We thank the reviewer for these constructive comments. All images presented in Figure 3A were captured using the same magnification, as now clarified in the revised figure legend. We appreciate the reviewer's question regarding the localization of *klf2* and *id2b*. While we were unable to directly visualize both markers in the same embryos due to the current unavailability of *klf2* reporter lines, prior studies using *klf2a:H2B-eGFP* transgenic zebrafish have demonstrated that *klf2a* is broadly expressed in endocardial cells, with enhanced expression in the atrioventricular canal region (Heckel et al., Curr Bio 2015, PMID: 25959969; Gálvez-Santisteban et al., Elife 2019, PMID: 31237233). Our *id2b:eGFP* reporter analysis revealed a similarly broad endocardial expression pattern. These independent observations support the likelihood that *klf2a* and *id2b* are co-expressed in the same endocardial cell population.

We also appreciate the reviewer's comments regarding the connection between biomechanical signaling and *id2b* expression. Previous studies have already established that biomechanical cues directly regulate *klf2* expression in zebrafish endocardial cells (Vermot et al., Plos Biol 2009, PMID: 19924233; Heckel et al., Curr Bio 2015, PMID: 25959969). In the present study, we observed a significant reduction in *id2b* expression in both *klf2a* and *klf2b* mutants, suggesting that *id2b* acts downstream of *klf2*. These observations together establish the role of biomechanical cues-*klf2*-*id2b* signaling axis in endocardial cells. Nevertheless, we agree with the reviewer that further investigation is required to elucidate the precise mechanism by which *klf2* regulates *id2b* expression.

> *(6) In Fig. 4: What's the mRNA expression for id2b in WT and id2b mutant fish hearts?*

We performed qRT-PCR analysis on purified zebrafish hearts and observed a significant reduction in *id2b* mRNA levels in *id2b* mutants compared to wild-type controls. These new results have been incorporated into the revised manuscript (Figure 4A).

> *(7) In Fig. 5E, the heart rate shows no difference between id2b+/+ and id2b-/- fish according to echocardiography analysis. However, Fig. 5B indicates a difference in heart rate. Could the authors explain this discrepancy?*

We thank the reviewer for this insightful observation. In our study, we observed a reduction in heart rate in *id2b* mutants during embryonic stages (120 hpf), as shown in Figure 5B. However, this difference was not evident in adult fish based on echocardiography analysis (Figure 5E). While the exact reason for these changes during development remains unclear, it is possible that the reduction in cardiac output observed in *id2b* mutants during early development triggers compensatory mechanisms over time, ultimately restoring heart rate in adulthood. Given that heart rate is primarily regulated by pacemaker activity, further investigation will be required to determine whether such compensatory adaptations occur and to elucidate the underlying mechanisms.

> *(8) In Fig. 6A: it's a little hard to read the gene names in the left most image in the panel. In Fig. 6B, the authors conducted qRT-PCR analysis of 72 hpf embryonic hearts and validated decreased nrg1 levels in id2b-/- compared to control. Since nrg1 is not specifically expressed in endocardial cells in the developing heart, the authors should isolate endocardial cells and compare nrg1 expression in id2b-/- to control. This would ensure that the loss of id2b affects nrg1 expression derived from endocardial cells rather than other cell types. In Supp Figure S6: Suggest adding an image of the UMAP projection to show tcf3b expression in endocardial cells from sequencing analysis.*

We thank the reviewer for these helpful suggestions. In response, we have increased the font size of gene names in the leftmost panel of Figure 6A to improve readability. Regarding *nrg1* expression, we acknowledge the importance of assessing its cell-type specificity.

Unfortunately, due to the lack of reliable transgenic or knock-in tools for *nrg1*, its precise expression pattern in embryonic hearts remains unclear. We attempted to isolate endocardial cells from embryonic hearts using FACS, but the limited number of cells obtained at this stage precluded reliable qRT-PCR analysis. Nonetheless, our data show that *id2b* is specifically expressed in endocardial cells, and publicly available single-cell RNA-seq datasets also support that *nrg1* is predominantly expressed in endocardial, but not myocardial or epicardial cells during embryonic heart development (Figure 6-figure supplement 1). These findings suggest that *id2b* may regulate *nrg1* expression in a cell-autonomous manner within the endocardium. As suggested, we have also added a UMAP image to Figure 7-figure supplement 1 to show *tcf3b* expression in endocardial cells, further supporting the cell identity in single-cell dataset.

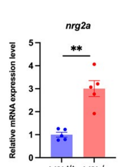
> *(9) In Fig. 6, Nrg1 knockout shows no gross morphological defects and normal trabeculation in larvae. Could the authors explain why they propose that endocardial id2b promotes nrg1 synthesis, thereby enhancing cardiomyocyte contractile function? Did Nrg1 knockdown with Mo lead to compromised calcium signaling and cardiac contractile function? Nrg2a has been reported to be expressed in endocardial cells in larvae, and its loss leads to heart function defects. Perhaps Nrg2a plays a more important role than Nrg1.*

We thank the reviewer for raising this important point. Although we did not directly test *nrg1* knockout in our study, previous reports have shown that genetic deletion of *nrg1* in zebrafish does not impair cardiac trabeculation during embryonic stages (Rasouli et al., Nat Commun 2017, PMID: 28485381; Brown et al., J Cell Mol Med 2018, PMID: 29265764). However, reduced trabecular area and signs of arrhythmia were observed in juvenile and adult fish (Brown et al., J Cell Mol Med 2018, PMID: 29265764), suggesting a potential role for *nrg1* in maintaining cardiac structure and function later in development. Whether calcium signaling and cardiac contractility are affected at these stages remains to be determined. Given that morpholino-induced knockdown is limited to early embryonic stages, it is not suitable for assessing *nrg1* function in juvenile or adult hearts.

As noted by the reviewer, *nrg2a* is expressed in endocardial cells, and its deletion has been associated with cardiac defects (Rasouli et al., Nat Commun 2017, PMID: 28485381). To assess its potential involvement in our model, we performed qRT-PCR analysis and observed increased *nrg2a* expression in *id2b* mutant hearts (Author response image 1). This upregulation may reflect a compensatory response to the loss of *id2b*. Therefore, *nrg2a* is unlikely to play an essential role in mediating the depressed cardiac function in this context.

Author response image 1.

Expression levels of *nrg2a*. qRT-PCR analysis of *nrg2a* mRNA in *id2b*$^{+/+}$ and *id2b*$^{-/-}$ adult hearts. Data were normalized to the expression of *actb1*. N=5 biological replicates, with each sample containing two adult hearts.



> *(10) In Fig. 7A of the IP experiment, it is recommended that the authors establish a negative control using control IgG corresponding to the primary antibody source. This control helps to differentiate non-specific background signal from specific antibody signal.*

As suggested, we have included an IgG control corresponding to the primary antibody species in the immunoprecipitation (IP) experiment to distinguish specific from non-specific binding. The updated data are presented in Figure 7A of the revised manuscript.

> *(11) In Pg. 5, line 115: there is no reference included for previous literature on blebbistatin.*

We have added the corresponding reference (Line 126, Reference #5).

> *In Pg. 5, lines 118-119; pg. 6 line 144: It would be beneficial to include a short sentence describing why choosing a tnnt2a morpholino knockdown to help provide mechanistic context to readers.*

We thank the reviewer for the constructive suggestion. In cardiomyocytes, *tnnt2a* encodes a sarcomeric protein essential for cardiac contraction, and its knockdown is a well-established method for abolishing heartbeat and blood flow in zebrafish embryos, thereby allowing investigation of flow-dependent gene regulation. In the revised manuscript, we have added a sentence and corresponding reference to clarify the rationale for using *tnnt2a* morpholino in our study (Lines 128-129, Reference #35).

> *In Pg. 6, line 140: Results title of "Cardiac contraction promotes endocardial id2b expression through primary cilia but not BMP" is misleading and contradicts the results presented in this section and corresponding figure. For example, the bmp Mo knockdown experiments led to decreased id2b fluorescence and the last statement of this results section contradicts the title that BMP does not promote endocardial id2b in lines 179-180: "Collectively, these results suggest that BMP signaling and blood flow modulate id2b expression in a developmental-stage-dependent manner." It would be helpful to clarify whether BMP signaling is involved in id2b expression or not.*

We apologize for any confusion caused by the section title. Our results demonstrate that *id2b* expression is regulated by both BMP signaling and biomechanical forces in a developmental-stage-specific manner. Specifically, morpholino-mediated knockdown of *bmp2b*, *bmp4*, and *bmp7a* at the 1-cell stage significantly reduced *id2b:eGFP* fluorescence at 24 hpf (Figure 3-figure supplement 1A, B), suggesting that *id2b* is responsive to BMP signaling during early embryonic development. However, treatment with the BMP inhibitor Dorsomorphin during later stages (24-48 or 36-60 hpf) did not significantly alter *id2b:eGFP* fluorescence intensity in individual endocardial cells, although a modest reduction in total endocardial cell number was noted (Figure 3-figure supplement 1C, D). These results suggest that BMP signaling is required for *id2b* expression during early development but becomes dispensable at later stages, when biomechanical cues may play a more prominent role. To address this concern and better reflect the data, we have revised the Results section title to: "BMP signaling and cardiac contraction regulate *id2b* expression". This revised title more accurately reflects the dual regulation of *id2b* expression (Line 153).

> *In line 205: Any speculation on why the hemodynamics was preserved between id2b mutant and WT siblings at 96 hpf?*

As suggested, we have included a sentence to address this observation. “Surprisingly, the pattern of hemodynamics was largely preserved in *id2b*$^{-/-}$ embryos compared to *id2b*$^{+/+}$ siblings at 96 hpf (Figure 4-figure supplement 1E, Video 1, 2), suggesting that the reduced number of endocardial cells in the AVC region was not sufficient to induce functional defects.” (Lines 223-225)

> *In line 246: Fig. 6k and 6j are referenced, but should be figure 5k and 5j.*

We have corrected this in the revised manuscript.

> ***Reviewer #2 (Recommendations for the authors):***
>
> *he manuscript was overall well explained, aside from a few minor points that would help facilitate reader comprehension:*
>
> *(1) The last paragraph of the introduction could be a brief summary of the study.*

We thank the reviewer for this constructive suggestion. As recommended, we have included a paragraph in the Introduction section summarizing our key findings to provide clearer context for the study (Lines 96-100).

> *(2) Lines 127-128: 'revealed a substantial recapitulation of the... of endogenous id2b expression' may need to be rephrased.*

We thank the reviewer for the valuable suggestion. In the revised manuscript, we have changed the sentence to: "Comparison of *id2b:eGFP* fluorescence with in situ hybridization at 24, 48, and 72 hpf revealed that the reporter signal closely recapitulates the endogenous *id2b* expression pattern." (Lines 137-139)

> *(3) Line 182: '... in a developmental-stage-dependent manner' sounds a bit ambiguous, may need to slightly elaborate/ clarify what this means.*

We thank the reviewer for the helpful comment. To improve clarity, we have revised the statement to: "Collectively, these results suggest that *id2b* expression is regulated by both BMP and biomechanical signaling, with the relative contribution of each pathway varying across developmental stages." (Lines 195-197)

> ***Reviewer #3 (Recommendations for the authors):***
>
> *(1) The conclusion that BMP signaling prior to 24 hpf is necessary for id2b expression is not fully supported by the data. How do the authors envision pre-linear heart tube BMP signaling impacting endocardial id2b expression during later chamber stages? Id2b reporter fluorescence can be clearly visualized in the linear heart tube in panel B from Figure 1. Does id2b expression initiate prior to contraction? Can the model be refined by showing when id2b endocardial reporter fluorescence is first observed, and whether this early/pre-contractile expression is dependent on BMP signaling?*

We thank the reviewer for the important comment. As suggested, we performed morpholino-mediated knockdown of *bmp2b*, *bmp4*, and *bmp7a* at the 1-cell stage. Live imaging at 24 hpf showed significantly reduced *id2b:eGFP* fluorescence compared to controls (Figure 3-figure supplement 1A, B), suggesting that *id2b* is responsive to BMP signaling during early embryonic development. However, treatment with the BMP inhibitor Dorsomorphin during 24-48 or 36-60 hpf did not significantly impact *id2b:eGFP* fluorescence intensity in individual endocardial cells, although a reduction in endocardial cell number was observed (Figure 3-figure supplement 1C, D). These results suggest that BMP signaling is essential for *id2b* expression during early embryonic development, while it becomes dispensable at later stages, when biomechanical cues exert a more significant role.

> *(2) Overexpressing tagged versions of TCF3b and Id2b in HEK293 cells is a very artificial way to make the major claim that these two proteins interact in endogenous endocardial*

> *cells. Can this be done in zebrafish embryonic or adult hearts?*

We thank the reviewer for this insightful comment. As suggested, we synthesized Flag-*id2b* and HA-*tcf3b* mRNA and co-injected them into 1-cell stage zebrafish embryos. We collected 100-300 embryos at 12, 24, and 48 hpf and performed western blot analysis using the same anti-HA and anti-Flag antibodies validated in HEK293 cell experiments. Despite multiple independent attempts, we were unable to detect clear bands of the tagged proteins in zebrafish embryos. We speculate that this could be due to mRNA instability, translational efficiency, or the low abundance of Id2b and Tcf3b proteins. We have acknowledged these technical limitations in the revised manuscript and clarified that the HEK293 cell data support a potential interaction between Id2b and Tcf3b, while confirming their endogenous interaction will require further investigations (Lines 295-296).

> *(3) The data presented are consistent with the claim that the tcf3b binding sites are functional upstream of nrg1 to repress its transcription. To fully support this idea, those two sites should be disrupted with gRNAs if possible.*

We thank the reviewer for the valuable suggestion. In response, we attempted to disrupt the *tcf3b* binding sites using sgRNAs. However, we encountered technical difficulties in identifying sgRNAs that specifically and efficiently target these binding sites without affecting adjacent regions. Despite these challenges, our luciferase reporter assay, using *tcf3b* mRNA overexpression and morpholino knockdown, clearly demonstrated that *tcf3b* binds to and regulates *nrg1* promoter region. Nevertheless, we acknowledge that future study using genome editing will be necessary to validate the direct binding of *tcf3b* to *nrg1* promoter.

> *Minor Points:*
>
> *(1) Must remove all of the "data not shown" statements and add the primary data to the Supplemental Figures.*

As suggested, we have removed all of the “data not shown” statements and added the original data to the revised manuscript (Figure 4E, middle panels, and Figure 4-figure supplement 1F)

> *(2) Must present the order of the panels in the figure as they are presented in the text. One example is Figure 6 where 6E is discussed in the text before 6C and 6D.*

We thank the reviewer for bring up this important point. In the revised manuscript, we have carefully revised the manuscript to ensure that the order of figure panels matches the sequence in which they are discussed in the text. Specifically, we have reorganized the presentation of Figure 6 panels to align with the text flow, discussing panels 6C and 6D before panel 6E. The updated figure and corresponding text have been corrected accordingly in the revised manuscript.

> *(3) Change the italicized gene names (e.g. tcf3b) to non-italicized names with the first letter capitalized (e.g. Tcf3b) when referencing the protein.*

As suggested, we have revised the manuscript to use non-italicized names with the first letter capitalized when referring to proteins.

> *(4) All bar graphs should be replaced with dot bar graphs.*

We have replaced all bar graphs with dot bar graphs throughout the manuscript.

> *(5) The new id2b mutant allele should be validated as a true null using quantitative RT-PCR to show that the message becomes destabilized through non-sense mediated decay or by immunostaining/western blot analysis if there is a zebrafish Id2b-specific antibody available.*

We thank the reviewer for this important suggestion. We have performed qRT-PCR analysis and detected a significant reduction in *id2b* mRNA levels in *id2b*$^{-/-}$ compared to *id2b*$^{+/+}$ controls. These new results are presented in Figure 4A of the revised manuscript.

> *(6) Was tricaine used to anesthetize embryos for capturing heart rate and percent fractional area change? This analysis should be performed with no or very limited tricaine as it affects heart rate and systolic function. These parameters were captured at 120 hpf, but the authors should also look earlier at 72 hpf at a time when valves are not present by calcium transients are necessary to support heart function.*

We thank the reviewer for this important comment. When performing live imaging to assess cardiac contractile function, we used low-dose tricaine (0.16 mg/mL) to anesthetize the zebrafish embryos. We have included this important information in the Methods section (Line 503). As suggested, we have also included the heart function results at 72 hpf, which are now presented in Figure 5-figure supplement 2A-C of the revised manuscript.

> *(7) The alpha-actinin staining in Figure 5-supplement 2D is very pixelated and unconvincing. This should be repeated and imaged at a higher resolution.*

As suggested, we have re-performed the α-actinin staining and acquired higher-resolution images. The updated results are now presented in Figure 5-figure supplement 2G of the revised manuscript.

> *(8) The authors claim that reductions in id2b mutant heart contractility are due to perturbed calcium transients instead of sarcomere integrity. Why do the authors think that regulation of calcium dynamics was not observed in the DEG enriched GO-terms? Was significant downregulation of cacna1 identified in the bulk RNAseq?*

We thank the reviewer for raising this important point. In our bulk RNAseq dataset comparing *id2b* mutant and control hearts, GO term enrichment was primarily associated with pathways related to cardiac muscle contraction and heart contraction (Figure 5-figure supplement 1B). We speculate that the transcriptional changes related to calcium dynamics may be relatively subtle and thus were not captured as significantly enriched GO terms. In addition, our qRT-PCR analysis revealed a significant reduction in *cacna1c* expression in *id2b* mutant hearts compared to controls, suggesting that *id2b* deletion impairs calcium channel expression. However, this change was not detected by RNA-seq, likely due to limitations in sensitivity.

> *(9) In line 277, the authors say, "To determine whether this interaction occurs in zebrafish, Flag-id2b and HA-tcf3b were co-expressed in HEK293 cells...". This should be re-phrased to, "To determine if zebrafish Id2b and Tcf3b interact in vitro, Flag-id2b and HA-tcf3b were co-expressed in HEK293 cells for co-immunoprecipitation analysis." The sentence in line 275 should be changed to, "....heterodimer with Tcf3b to limit its function as a potent transcriptional repressor."*

We thank the reviewer for these constructive comments and have revised the text accordingly (Lines 291-294).

> *(10) Small text corrections or ideas:*
>
> *Line 63: emphasized*

We have corrected this in the revised manuscript.

> *Line 71: studied signaling pathways*

We have corrected this in the revised manuscript.

> *Line 106: the top 6 DEGS (I think that the authors mean top 6 GO-terms) and is Id2b in one of the enriched GO categories?*

id2b is one of the top DEGs. This point has been clarified in the revised manuscript (Lines 116-117).

> *Line 125: a knockin id2b:eGFP reporter line*

We have corrected this in the revised manuscript (Line 136).

> *Line 138: This paragraph could use a conclusion sentence.*

We have added a conclusion sentence in the revised manuscript (Lines 150-151).

> *Line 190: id2b-/- zebrafish experienced early lethality*

We have revised the statement as suggested (Line 206).

> *Line 193: The prominent enlargement of the atrium with a smaller ventricle has characterized as cardiomyopathy in zebrafish (Weeks et al. Cardiovasc Res, 2024, PMID: 38900908), which has also been associated with disruptions in calcium transients (Kamel et al J Cardiovasc Dev Dis, 2021, PMID: 33924051 and Kamel et al, Nat Commun 2021, PMID: 34887420). This information should be included in the text along with these references.*

We thank the reviewer for this helpful suggestion. We have incorporated these important references into the revised manuscript and included the relevant information to acknowledge the established link between atrial enlargement, cardiomyopathy, and disrupted calcium transients in zebrafish models (Reference #41, 42, and 45; Lines 210 and 260).

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